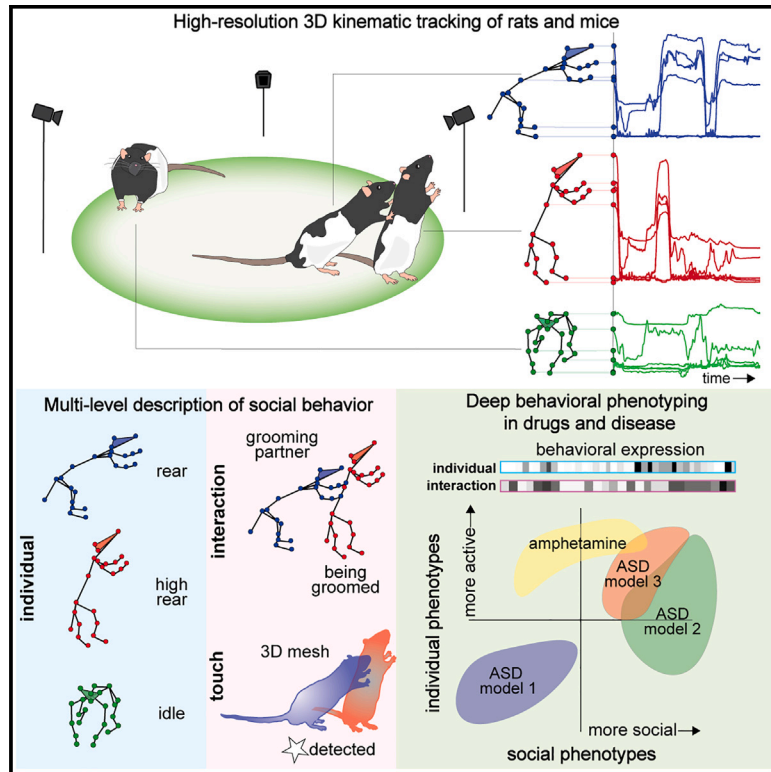


Mapping the landscape of social behavior

Graphical abstract



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In brief

Computer vision and machine learning approaches map a multi-scale social behavioral landscape, revealing ASD signatures and establishing an atlas of rodent sociality.

Highlights

- s-DANNCE uses machine learning to track and catalog social behavior in rats and mice
- It identified a rich landscape of actions, interactions, synchrony, and body contacts
- It revealed distinct social phenotypes of mouse strains and monogenic rat ASD models
- We release a large database of tracked interactions to advance social neuroscience



Resource

Mapping the landscape of social behavior

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SUMMARY

Social interaction is integral to animal behavior. However, lacking tools to describe it in quantitative and rigorous ways has limited our understanding of its structure, underlying principles, and the neuropsychiatric disorders, like autism, that perturb it. Here, we present a technique for high-resolution 3D tracking of postural dynamics and social touch in freely interacting animals, solving the challenging subject occlusion and part-assignment problems using 3D geometric reasoning, graph neural networks, and semi-supervised learning. We collected over 110 million 3D pose samples in interacting rats and mice, including seven monogenic autism rat lines. Using a multi-scale embedding approach, we identified a rich landscape of stereotyped actions, interactions, synchrony, and body contacts. This high-resolution phenotyping revealed a spectrum of changes in autism models and in response to amphetamine not resolved by conventional measurements. Our framework and large library of interactions will facilitate studies of social behaviors and their neurobiological underpinnings.

INTRODUCTION

The study of social interactions is essential for understanding evolutionary, ecological, and neurobiological principles of animal behavior. While social behavior is multifaceted, much of it is expressed through body movements that reflect internal states, goals, and inter-animal communication. These gestures are often subtle, necessitating a method for capturing the precise 3D kinematics of interacting animals. Such measurements could then be analyzed to define and quantify social behaviors in rigorous and reproducible ways, enabling sophisticated interrogations of social dynamics and their biological underpinnings. Further, an automated and scalable method would permit comprehensive behavioral screens to characterize the full variety of social phenotypes, including in autism and other neurodevelopmental disorders.

These goals have yet to be met. Today, social phenotyping is typically done by manually scoring video recordings or, increasingly, by automatically tracking the position and orientation of interacting animals. For instance, in studies of rodent models of autism, metrics of sociality are commonly derived from the fraction of time an animal spends in the proximity of a caged conspecific.¹ While some studies expand these metrics to include the occurrence of a limited set of social action types,² the manual annotation that goes into this is laborious, inconsistent, and limited to predefined categories. For autism and other conditions diagnosed using only behavioral criteria, such coarseness

makes it difficult to establish animal model face validity and to translate findings out of the lab.³

To reliably discern and describe subtle variations in social phenotypes,⁴ we need an approach that can capture the diversity and nuances of social behavior. That means precisely tracking how interacting animals move, synchronize, and make physical contact, plus analytics for classifying, interpreting, and comparing these elements of interaction. Capturing the nature of physical contact between social partners is particularly important given recent interest in social touch as a rewarding stimulus and its impairment in autism spectrum disorders (ASDs).^{5,6} Further, these analytics must address multiple timescales of social interactions, from short, stereotyped movements and engagements to behavioral patterns that evolve over longer times, such as group coordination, instigation, communication, and arousal.⁷

Recent advances in video-based behavioral quantification have enabled more granular and scalable measurements of social interactions, but these methods remain limited in resolution. Instead of the coarse position and orientation tracking afforded by classical computer vision, convolutional neural networks (CNNs) can now track anatomical keypoints of interacting animals in 2D.^{8,9} With these 2D kinematic descriptions, classifiers can be trained to detect different types of social interactions once they are enumerated and labeled by humans.¹⁰ Tracked 2D kinematics can also be clustered without explicit human supervision, facilitating the discovery and annotation of social



action patterns not captured by human-defined behavioral categories.^{11–13} Nevertheless, 2D measurements are inherently limited in terms of the body parts that can be tracked, and perspective ambiguities make it difficult to derive reliable body kinematics from a single view. Thus, while these 2D innovations are significant, they are not yet precise enough to provide comprehensive descriptions of social behavior.

Reliable quantitative descriptions of social interactions would be greatly facilitated by high-resolution 3D tracking of animal pose—that is, the positions of actuatable body parts, including limbs—which would permit precise spatiotemporal profiling of coordinated kinematics and body contact. While such 3D pose tracking methods have been developed for single animals,^{14–19} extending them to social contexts is far from trivial. Single-animal methods are not equipped to deal with the complexities of multi-animal environments, where animal identities must be tracked consistently and bodies are often hidden from view. Depth imaging has been used for social tracking but cannot faithfully track limb movement.^{20–22} Tracking animals as single 3D points within groups has been foundational for studies of social and collective behavior^{23,24} but cannot and does not aim to measure detailed body movements.²⁵ 2D whole-body pose tracking methods could, in principle, be extended to 3D via triangulation across multiple camera viewpoints,²⁶ but inevitable and ubiquitous animal-animal occlusions make 2D-to-3D triangulation during interaction particularly challenging. Large marker-based 3D training datasets can be used to address occlusion difficulties, but markers themselves are often occluded on limbs and during close interaction, limiting these datasets to reduced keypoint sets on the head and trunk.^{27–29} Thus, reliable whole-body 3D tracking of social behavior has yet to be achieved.

To address these problems, we developed a method for tracking highly resolved 3D postural kinematics in interacting animals using commercially available video cameras. Our approach, social-DANNCE (s-DANNCE), builds off DANNCE, a deep neural network for high-resolution 3D markerless tracking in single animals.¹⁷ s-DANNCE leverages semi-supervised learning, anatomical constraints, and a graph neural network (GNN) to resolve animal-animal occlusions. It reliably tracks pairs of interacting rats with performance surpassing previous approaches^{26,27} and rivaling that of human labelers. s-DANNCE is flexible, supporting measurements across a range of experimental settings, including in larger groups of animals, and generalizes across species from rats to mice.

To parse high-resolution tracking data into comprehensive and quantitative descriptions of social behavior across spatio-temporal scales, we developed a suite of computational tools for analyzing dyadic interactions (Figure 1A), including the automatic identification of recurring and stereotyped social interaction motifs. By fitting a volumetric body model to 3D poses, we also measured social touch, providing quantitative access to a salient mode of social exchange.³⁰

To validate whether our pipeline enhanced social phenotyping, we gave rats amphetamine, a drug with clear and demonstrable social effects in humans but inconsistent social effects in rodents.^{31,32} In addition to inducing overall hyperactivity as expected, drug administration disrupted behavioral synchrony,

modulated canonical interaction motifs, and changed social touch distributions across animal bodies. Social phenotyping was also enhanced in mice, where expanded behavioral dimensions distinguished strains used as models of low and high sociability.

Our approach revealed subtle deviations from typical social behavior in seven rat models of autism. We found rats from each model were impacted distinctively, with four models demonstrating widespread shifts in social behavior, establishing distinct candidate models of face-valid social phenotypes. Beyond advancing the state of the art in social tracking, our study produced a large database of 140 million rat and mouse 3D pose samples in lone and social contexts. We provide these data, together with our machine learning framework and detailed phenotyping results, as a resource for the community to mine, model, and reference in the pursuit of a deeper understanding of social behavior.

RESULTS

Animal-animal occlusions challenge 3D pose tracking during social interaction

The state of the art for quantifying animal movement during social behaviors is to track 2D anatomical features, or keypoints, on interacting subjects, typically from videos recorded by a single top-down or bottom-up camera.^{8–10} However, quantifying behavior in 2D pixel space, instead of descriptions of true 3D movements (in millimeters), introduces kinematic artifacts due to perspective and depth ambiguities.^{15,33} Single-camera measurements are also sensitive to body occlusions, which increase during social interaction (Figures S1B and S1C). In a top-down view, the animal's own body occludes its appendages 76% of the time, while occluding the head and trunk for 5% and 10% of the time, respectively (Figure S1D). In a bottom-up view, the top of the body is always occluded, as are the forelimbs and head in elevated postures, such as rearing (Figure S1E). During close social interaction, one animal often obstructs the view of the other, increasing occlusions (3.0- and 1.5-fold more head and trunk occlusions, respectively, than a lone animal; Figures S1D and S1E).

In principle, such occlusions and kinematic artifacts can be addressed via 3D tracking of body keypoints in multicamera videos, as done in single animals.^{14–17,27} To test this, we built an arena setup with six synchronized 50 Hz color video cameras and recorded rat dyads (Figures 1B and S1A; Video S1). A previous study applied DANNCE, a 3D tracking method using volumetric neural networks, to multiple animals.^{17,27} This multi-animal implementation exploited DANNCE's "top-down" structure, where animals are identified and compartmentalized into 3D bounding boxes before pose estimation. However, while DANNCE reliably tracked 3D pose when animals were far apart, performance degraded significantly during close interactions. For example, we frequently observed "chimeric" errors, where the inferred 3D pose incorrectly integrated body elements from its interaction partner (Figure S2). This failure mode arises from animals being enclosed, at least partially, in the same 3D volume, producing multiple candidate keypoint locations for individual body parts (Figure 1B, bottom).

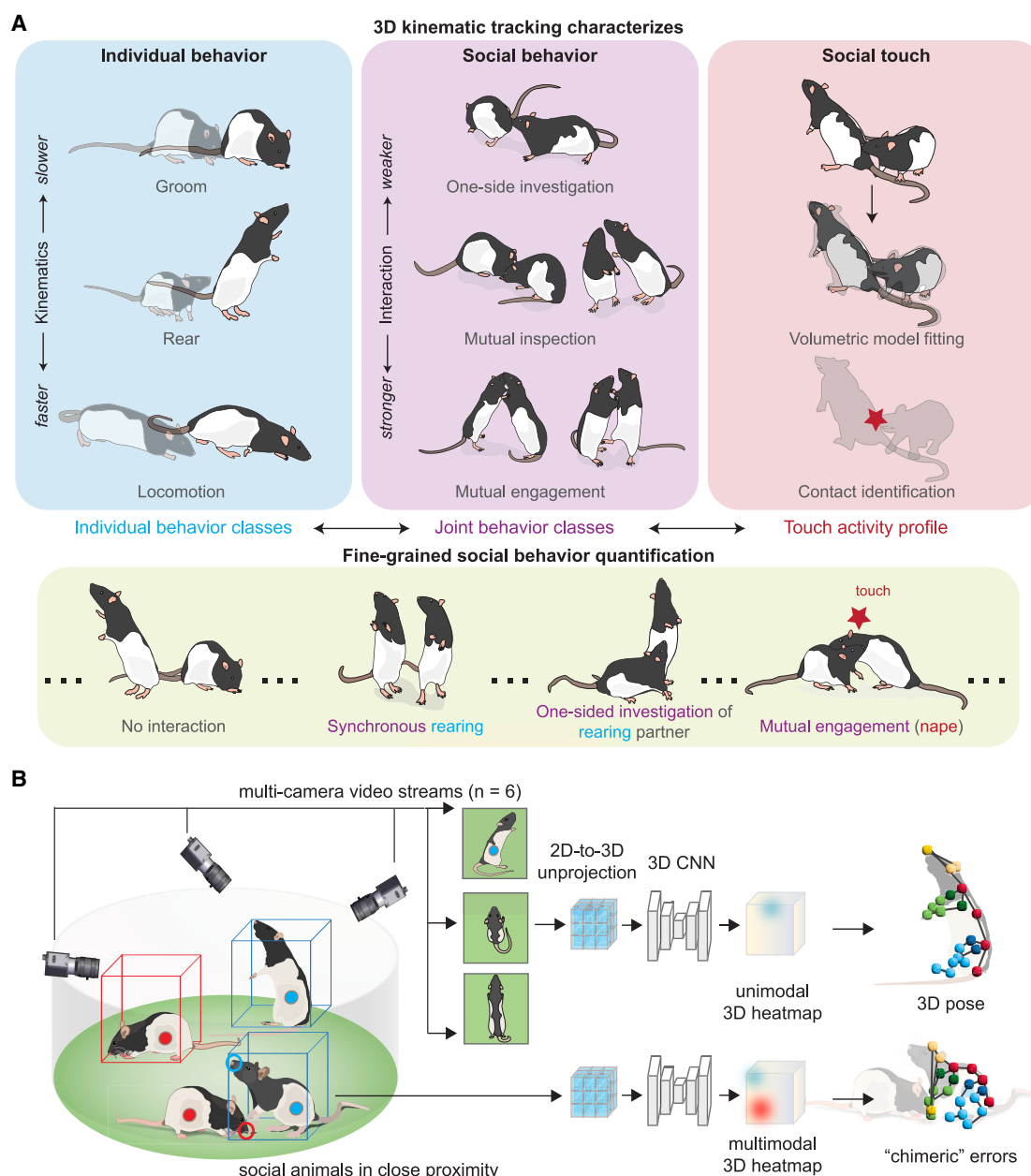


Figure 1. High-resolution 3D kinematic tracking enables fine-grained quantification of social behaviors

(A) We introduce three levels of behavioral profiling based on 3D keypoint tracking: individual behaviors, social interaction behaviors, and social touch. These enable fine-grained curation and quantification of social behaviors in freely interacting animals.

(B) Schematic of recording arena and baseline DANNCE implementation, where pose estimation is performed separately within 3D volumes encapsulating each animal. During close interactions, keypoints from multiple animals overlap inside volumes (open red and blue circles), producing multimodal 3D heatmaps and chimeric errors.

Related to Figure S1.

s-DANNCE enables 3D tracking of social behaviors

To overcome the challenges of consistently and precisely tracking whole-body 3D kinematics during social behaviors, we developed s-DANNCE, which extends the precision of volumetric deep neural networks to closely interacting animals (Figures 2A and S2A). s-DANNCE combats pervasive chimeric

prediction errors and other observed anatomical inconsistencies (collapsed poses and errant body segment lengths) by imposing explicit constraints on anatomical structure during training, plus a GNN module³⁴ enabling skeleton-aware 3D image feature processing. Further, s-DANNCE is able to leverage both labeled and unlabeled data during training, allowing the

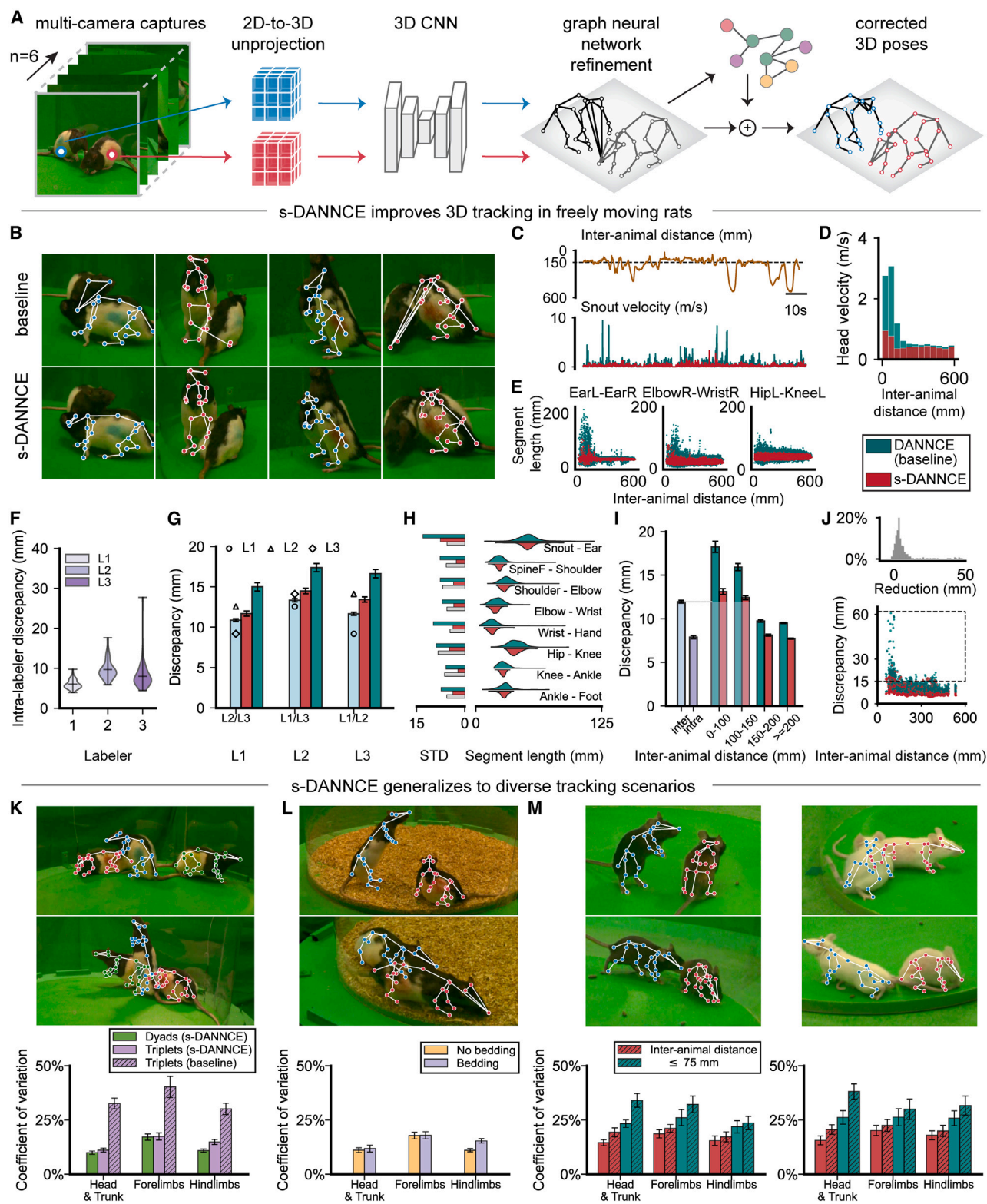


Figure 2. s-DANNCE improves 3D tracking of freely interacting animals

(A) Schematic of the s-DANNCE pipeline (cf. Figure S2A).

(B) Examples (projected 3D poses) of baseline DANNCE and s-DANNCE predictions. Red and blue denote separate animals.

(legend continued on next page)

network to learn from large, diverse datasets not annotated by users.^{19,35}

To test whether s-DANNCE mitigated naive DANNCE baseline tracking issues, we trained both on the same frames of freely interacting rat pairs ($n = 423$ manually annotated 23-keypoint 3D poses and $n = 910$ unlabeled frames) and tested the models on videos of animals not used for training. We found that s-DANNCE, in contrast to baseline DANNCE, produced robust tracking and resolved chimeric errors (Figures 2B, S2D–S2H, and S3). Baseline DANNCE exhibited large fluctuations in keypoint velocities (Figures 2C and 2D) and body segment lengths (Figure 2E) when animals were close to each other (<150 mm inter-animal distance, $\sim 2/3$ rat body length), indicating frequent social tracking errors. These errors were resolved by s-DANNCE (65% and 60% improvement in head velocity and limb length stability, respectively, for s-DANNCE over DANNCE; Figures 2D and 2E).

To more directly assess s-DANNCE performance, we quantified how close tracked 3D keypoints were to human annotations in close social interaction frames. We had three labelers annotate the same test set for these comparisons ($n = 80$ unique 3D poses, with poses annotated twice by each labeler for measuring intralabeler consistency), allowing us to account for human labeling biases (Figure 2F). s-DANNCE rivaled human labeler precision, with it placing keypoints closer to human annotations than one of the labelers in the group and nearing interlabeler precision on average (Figure 2G; s-DANNCE 13.20 ± 0.19 mm discrepancy relative to all labelers; interlabeler 11.95 ± 0.16 mm discrepancy over all labeler pairs; mean $\pm$ 95% CI). By contrast, baseline DANNCE produced consistently larger discrepancies relative to human annotators (16.34 ± 0.29 mm, Figure 2G). A recent 3D social tracking approach using keypoint triangulation with image masking²⁶ also yielded larger discrepancies (24.64 ± 0.36 mm), which persisted even with manually annotated image masks (18.88 ± 0.24 mm, Figures S3K–S3H). 3D poses by s-DANNCE were also more anatomically plausible than those of human labelers, as evidenced by consistent body segment lengths across postures (s-DANNCE 0.14 ± 0.022 , human labeler 0.17 ± 0.019 , baseline DANNCE 0.30 ± 0.057 , coefficient of variation $\pm$ 95% CI, Figure 2H). We further examined the performance of s-DANNCE over a broader variety of social interactions in a larger test recording ($n = 2,746$ 3D poses in one pair of rats, annotated by labeler 1). The largest improvements were found during the most

interactive social behaviors when animals were close and occluding one another. Specifically, s-DANNCE yielded 6.7-fold better precision relative to baseline DANNCE when adjusting for human inter-labeling uncertainty (0.72 ± 0.20 vs. 4.84 ± 0.36 mm, Figure 2I; 12.67 ± 0.19 vs. 16.80 ± 0.35 mm without adjustments). s-DANNCE outperformed baseline DANNCE in 94% of test frames, and the remaining 6% were within the interlabeler margin of error (Figure 2J).

s-DANNCE extended to a wide range of experimental contexts, including recordings of interacting mice. While training frames came from just one strain of rats recorded in one arena, s-DANNCE generalized to unseen arenas, rat strains, and larger social groups without additional pose annotations (Figures 2K, S3A–S3D, and S3G–S3J). Adding only a small number annotated frames for training, s-DANNCE could accurately track social behavior in arenas with bedding (Figures 2L and S3H) and in experiments with two different and visually distinct strains of freely interacting mice (Figures 2M, S3E, and S3F). Our approach thus supports high-resolution quantification of social behavior across a broad range of conditions with limited additional annotations.

3D kinematic profiling reveals multi-scale social phenotypes

High-resolution and continuous whole-body 3D kinematic measurements provide an opportunity to quantitatively map social behavioral repertoires across animals and experimental conditions. To characterize social behavior, we first examined how individual repertoires changed in lone vs. social contexts using a method that clusters behavior into stereotyped motifs based on an individual subject's 3D postural dynamics.^{36–38} Briefly, for each movie, we derived a time series of high-dimensional features from 3D pose dynamics and applied nonlinear dimensionality reduction to generate a 2D behavior map. We performed spatial clustering of this map to identify stereotyped action classes, which were used to assign a behavioral label at each tracked time point (Figures 3A and 3E). We manually reviewed and annotated behavioral snippets sampled from each cluster on the behavior map, revealing an action space organized into regions representing 9 human-annotated high-level action classes (HLACs) and 162 low-level action clusters (LLACs) distinguishable by 3D kinematics (Figures S4A–S4C; Table S1; Videos S3 and S4; STAR Methods). This allowed us to compare behavior at the level of fine stereotyped actions (LLACs) or with coarser, more interpretable

(C–E) Data from a test video sequence ($n = 90,000$ frames) in a rat engaged in dyadic interactions. Throughout this figure, green denotes the baseline DANNCE, and red denotes the s-DANNCE model. (C) Traces (80 s) of inter-animal distance (top) and corresponding snout velocity (bottom). (D) 95th percentile of head velocity within 50 mm bins of inter-animal distance. (E) Scatter plots of indicated body segment lengths vs. inter-animal distance. (F–H) Performance relative to three human labelers (L1, L2, and L3) during close interactions. Each annotated the same $n = 40$ frames from $n = 4$ animals ($n = 1,840$ keypoints per labeler), twice. (F) Violin plots of the intralabeler annotation discrepancies, showing means and extrema. (G) Bar plots of discrepancies relative to each labeler. (H) Left: segment length standard deviations (SD) for model predictions and human annotations (L1, L2, and L3 mean). Right: distributions of model-derived body segment lengths for the labeled frames. Laterally symmetric body segments are merged. (I) Keypoint localization discrepancy over a larger dataset from different animals, annotated by L1 ($n = 1,373$ frames from $n = 2$ animals; $n = 2,746$ poses, 63,158 keypoints), binned by inter-animal distance. Average intra- and interlabeler discrepancies from (F) and (G) are included for comparison. (G and I) error bars are 95% CIs across keypoints. (J) Bottom: scatter plot of discrepancy vs. inter-animal distance for the samples in (I). Top: histogram of s-DANNCE discrepancy reduction, for frames where baseline DANNCE discrepancies exceeded 15 mm (box in bottom, $n = 261$ poses). (K–M) (Top) s-DANNCE prediction examples (projected 3D poses) and (bottom) segment length coefficients of variation ($n = 12$ animals for each condition) for rat triplets (K), rat dyads with bedding (L), and two mouse strains (M). (K–M) error bars are 95% CIs across animals. Related to Figures S2 and S3.

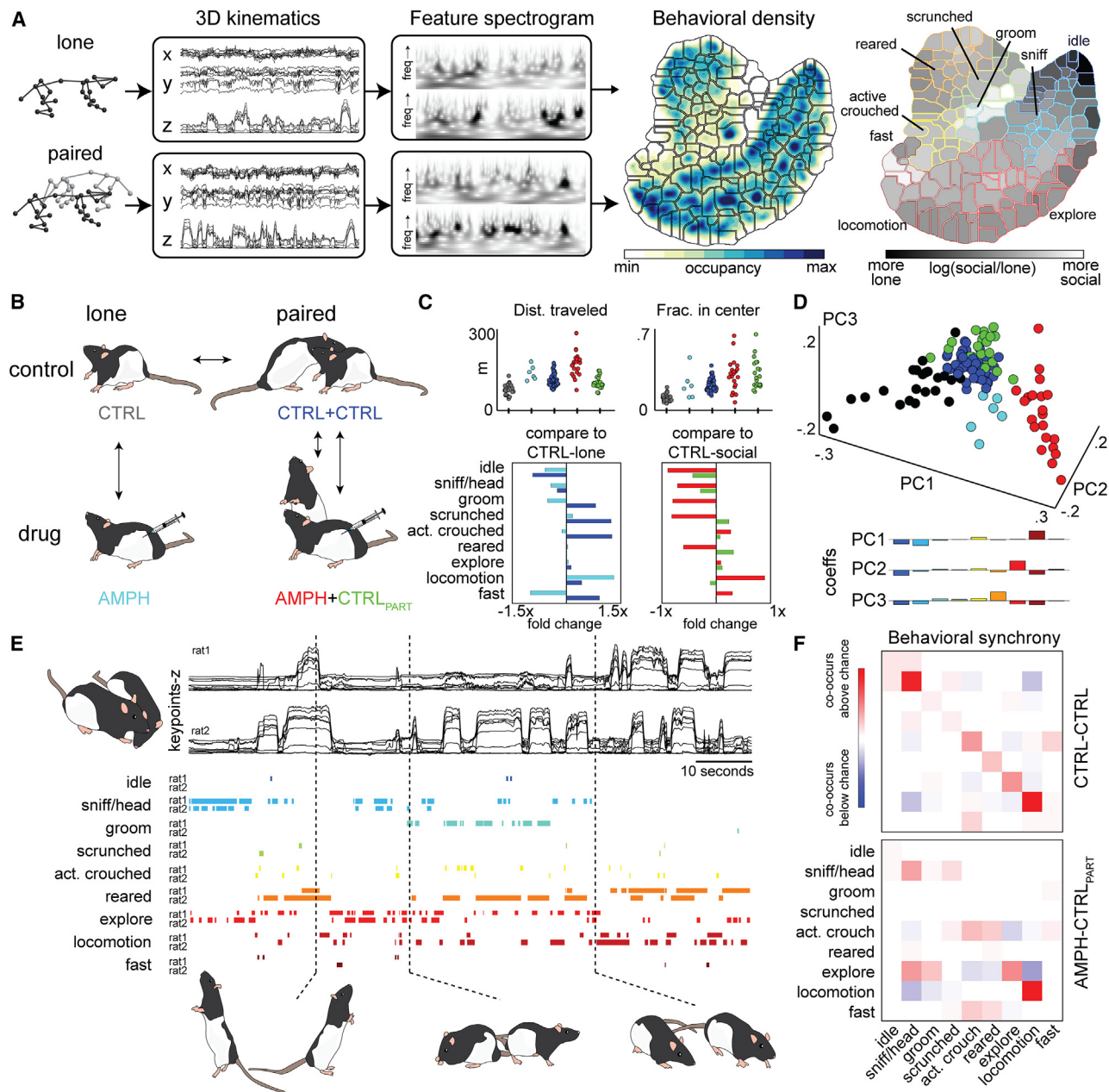


Figure 3. Spatiotemporal clustering identifies behavioral responses to social context and stimulant administration

(A) Pipeline transforming s-DANNCE poses to LLACs (outlined map regions) and HLACs (outline colors on right). The same color code is used in (D) (bottom) and (E).

(B) Schematic of behavioral contexts tested. Group label colors are reused for markers and bars in (C) and (D).

(C) Top: distance traveled and fraction of time in arena center over each 30-min recording, calculated from tracked animal center-of-mass (COM). Bottom: changes in HLAC occupancy across experimental groups.

(D) Scatter plot of PC scores (top) and coefficients (bottom) of HLAC occupancy for all recordings.

(E) Example keypoint kinematics and corresponding identified HLACs for CTRL+CTRL.

(F) HLAC partial mutual information (PMI) heatmaps for all recordings in the indicated groups.

operational definitions (HLACs). When we mapped full-body 3D kinematics from both single- and multi-animal recordings from wild-type (WT) Long Evans male rats ($n = 6$ animals, $n = 24$ pairings, $n = 24$ lone recordings) into the behavioral space, we

captured the behavioral shift associated with social context, revealing 116 LLACs (of 162 total) that exhibited significant changes in how often they were expressed (their behavioral “occupancies”) during social sessions (Figure 3A, right).

These quantitative descriptions enable comprehensive, unbiased, and automated ways of characterizing how drugs and diseases affect social behavior. To test the power of this approach, we tracked and profiled rat behavior in lone and social contexts with and without amphetamine administration ($n = 75$ recordings across social pairings and experimental conditions from $n = 6$ animals; total $n = 10.8$ million 3D poses) (Figure 3B; Video S2). Past studies, relying on coarse distance metrics or human annotation of broad behavioral categories, have been inconsistent, with some reporting no effect or effects specific to social play in juveniles,^{31,39} and have not dissociated social effects from amphetamine's substantial general effects on behavior.^{40,41} Our quantitative profiling automatically identified a variety of behavioral changes in response to amphetamine that were specific to social contexts and not readily detectable using coarser measures (Figures 3C and 3D). For instance, while amphetamine did not affect rearing behaviors in lone rats, in social situations it produced a pronounced decrease in frequency of rearing relative to controls (Figure 3C). Overall, large shifts in the behavioral repertoire were induced by the social context and further modulated by amphetamine. These shifts were consistent across animals and experimental sessions (Figure 3D; s-DANNCE repository⁴²). By contrast, conventional measures, such as distance traveled, were too coarse to distinguish between experimental groups (Figure 3C).

Identifying the HLACs expressed simultaneously by social partners allowed us to analyze behavioral synchrony (Figure 3F), a hallmark of social behavior that has been difficult to characterize or quantify without manual human annotation.^{43,44} We calculated how likely each HLAC combination was to co-occur between partners above what was expected by chance given each individual's HLAC occupancies (Figure 3F; STAR Methods). In WT pairings, the highest synchrony values were observed between behavioral classes of the same type (particularly both animals *sniffing* or both animals engaged in *locomotion*).

In addition to altering behavior expression frequencies in individual animals within social contexts (cf. Figure 3C), amphetamine administered to one animal in a social pairing reduced overall behavioral synchrony, although simultaneous locomotion was preserved. The behaviors expressed simultaneously when an amphetamine animal was present were often from different HLACs (Figure 3F). While these asymmetries could reflect one-sided increases in social engagement, common human responses to the drug,⁴⁵ determining this requires analysis of additional behavioral variables.

Dyadic embedding identifies stereotyped motifs of social interaction across species

While revealing, the kinematic profiles of individual animals do not capture the coordination of body movements unique to social interaction. To identify recurring and stereotyped social action motifs, we developed dyadic embedding, a classification method based on shared inter-animal features (Figures 4A and S4A–S4C). Our method uses the posture, dynamics, and inter-animal features from 3D tracking to produce a behavioral map that parses social behavior into stereotyped joint classes or motifs. The method effectively treats the two animals as a single system and exploits the fact that distinct social motifs manifest as identifiable

patterns in the relative positions and orientations of 3D poses between animals. For example, when one animal is sniffing and following the other from behind, the animals are consistently oriented in the same direction with heads maintained at a characteristic distance. Alternatively, when animals are investigating each other nose-to-nose, they are oriented in opposite directions with heads close together. We automatically discover each of these and numerous other social motifs from the joint embedding (STAR Methods). Crucially, social motifs should be sensitive to the egocentric perspectives of individual animals, preserving each individual's identity and context, as the meaning of an interaction is actor-dependent (Figures S4A–S4C).

Inspecting and annotating the behavioral clusters from multi-animal trajectories revealed highly specific stereotyped and recurring social interactions (Figure 4A; Video S4). Two axes of description were particularly informative in interpreting these clusters: the strength of, and the symmetry of, the interaction, each based on how much each animal appeared to engage in, or drive, the interaction (Figure 4B). Based on a manual review of movies sampled randomly from the 156 low-level joint behavior clusters (LLJCs), each cluster was assigned a descriptive label (e.g., “*rat1 following rat2, body length apart*” or “*both rearing, rat2 oriented toward rat1*”) (Table S1). We then assigned each LLJC to one of seven high-level joint behavior classes (HLJCs): *no interaction*, *lightly engaged*, *engaged*, *partner lightly engaged*, *light mutual interaction*, and *mutual interaction*. Together with the action clustering, this yielded a rich annotated social dataset where at each time point an individual is assigned to both an individual (animal-autonomous LLAC) and a joint (social interaction LLJC) behavioral cluster (Figure S4D; Video S5). Each of these low-level clusters also belongs to one of a smaller set of high-level classes (HLACs and HLJCs, respectively), providing multiple levels of description.

Using the dyadic embedding, we could describe in detail how amphetamine affects social behavior in rats (Figures 4B–4E). Our method revealed statistically significant effects that were undetectable using coarse metrics, such as center-of-mass movement or the distance between interacting animals (Figures 3C and 4C). When we compared HLAC occupancy between amphetamine and control conditions, we identified social behavioral types that were consistently modulated across the cohort (Figure 4E). Specifically, amphetamine increased social engagement and investigatory behaviors while reducing these behaviors in undosed social partners, compared with recordings where neither animal had received amphetamine (Figure 4C, bottom). There were more complex changes at the finer LLJC level, where, for example, different *engagement-related behaviors* were not uniformly upregulated. For instance, LLJCs for touching a rearing partner from the front or side exhibited several-fold increases after amphetamine administration (2.7- and 2.2-fold, respectively, Figures 4D and 4E), whereas investigating and grooming a partner from above exhibited the largest decrease (2.0-fold, Figures 4D and 4E). Some HLJCs, such as *mutual interaction*, appeared unaffected, but we found this was due to divergent responses in LLJCs comprising this high-level class (10 *mutual interaction* LLJCs significantly upregulated and 5 downregulated, Figure 4D; Table S1), reiterating the value of our high-resolution, multilevel analysis.

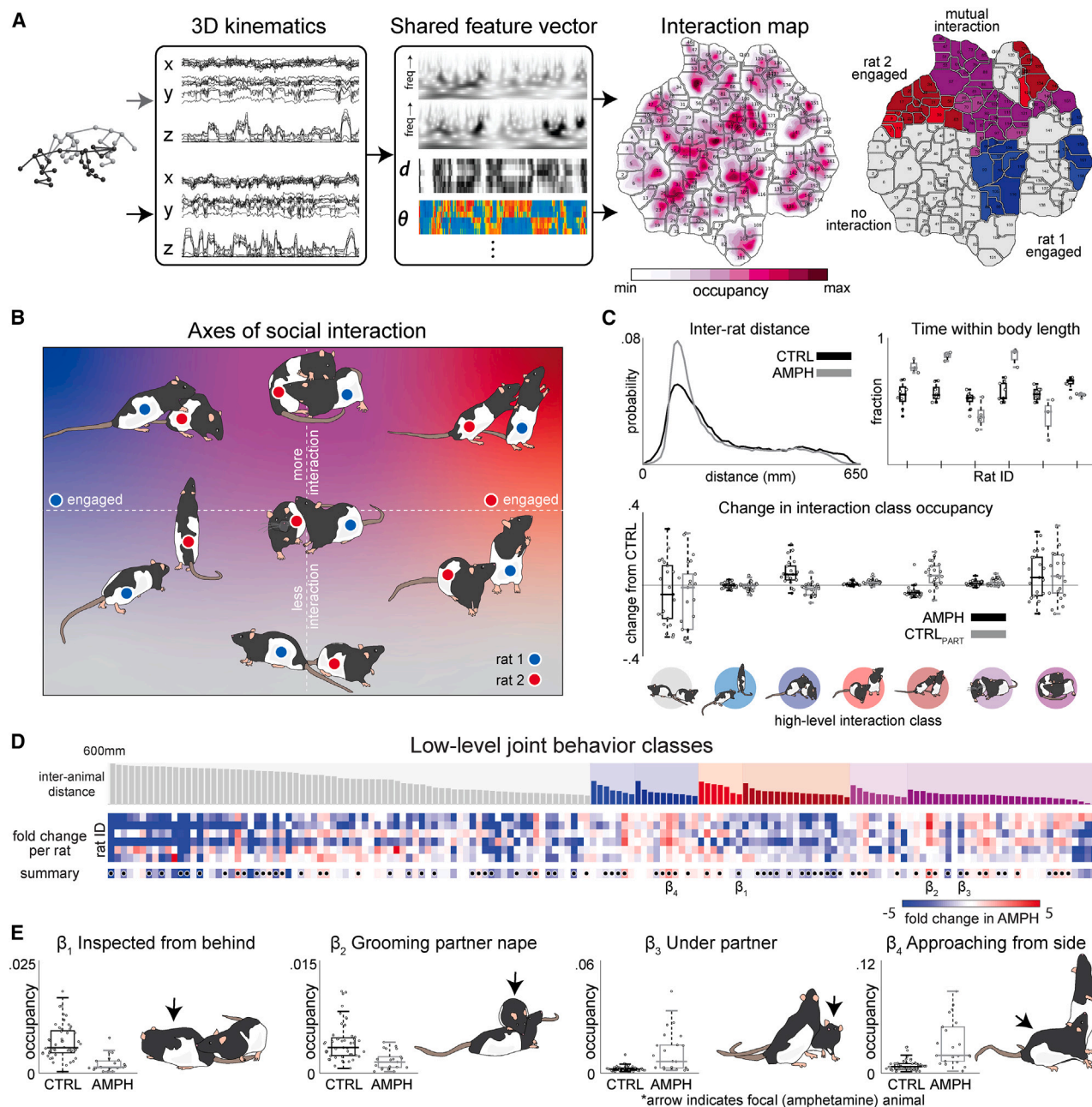


Figure 4. Dyadic feature embedding reveals amphetamine-associated social differences

(A) Pipeline transforming s-DANNCE poses to LLJCs (outlined map regions) and HLJCs (colored regions on right).
 (B) Snapshots of representative behaviors from the seven HLJCs. The color scheme is also used in (C), bottom, and (D), top: gray, no interaction; dark red, rat2 engaged; dark blue, rat1 engaged; purple, mutual interaction; lighter hues are HLJCs with lower interaction intensity.
 (C) Top left: inter-animal distance distribution across all animals in CTRL+CTRL and AMPH+CTRL_{PART} pairings. Top right: fraction of time within a body length to partner in AMPH+CTRL_{PART}, for each animal (black, when CTRL; gray, AMPH). Bottom: HLJC occupancy change across all AMPH+CTRL_{PART} recordings (dots) relative to CTRL+CTRL HLJC occupancy for each animal.
 (D) Heatmap of per-animal mean LLJC occupancy fold change for AMPH vs. CTRL_{PART} conditions and mean-fold change across animals (summary), sorted by HLJC and inter-animal distance. Dots $p < 0.05$ (mixed effects), Benjamini-Hochberg false discovery rate (BHFD) correction for multiple comparisons ([STAR Methods](#)).
 (E) Detailed boxplots for the indicated LLJCs.
 Related to [Figure S4](#).

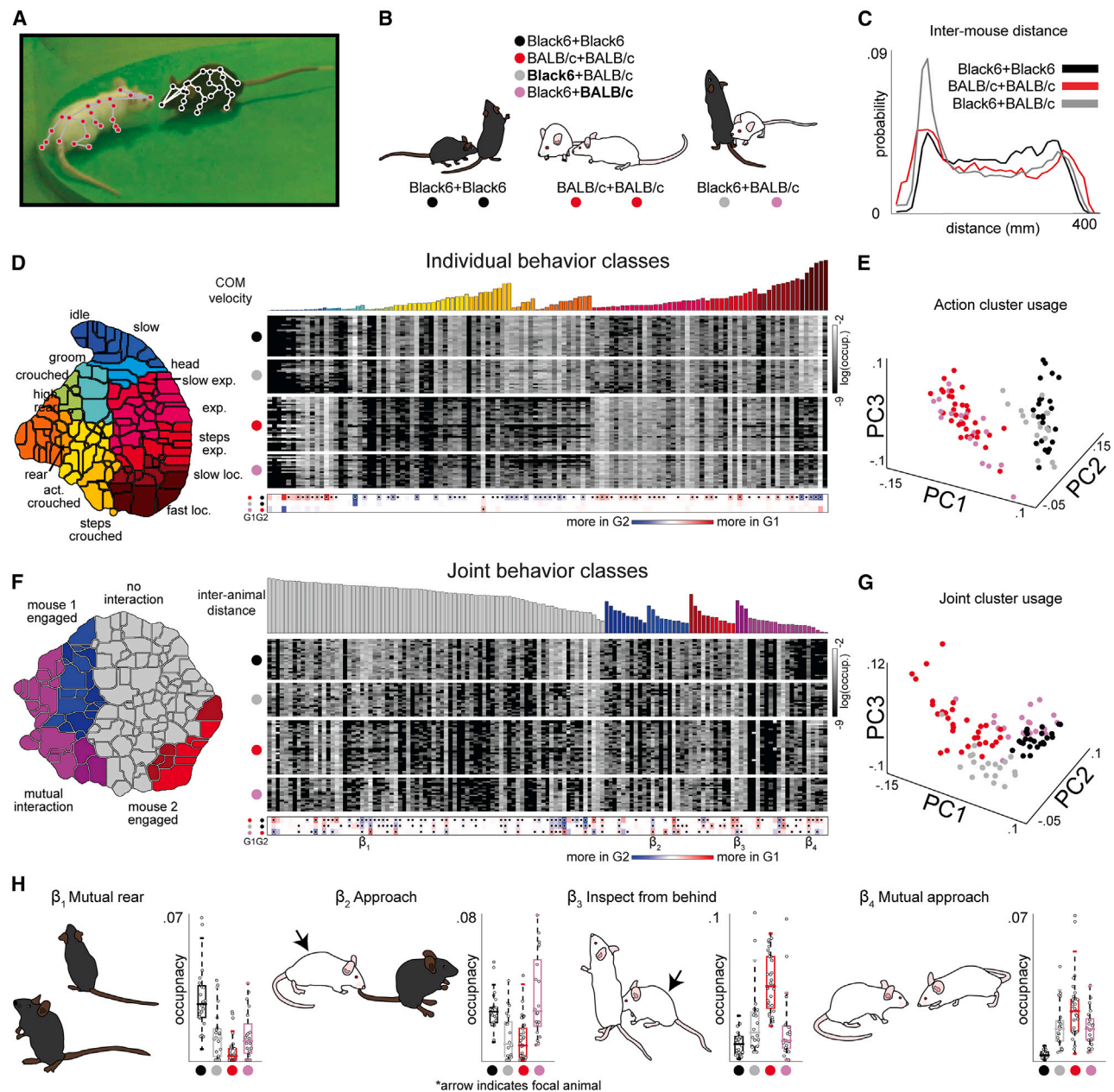


Figure 5. Mouse social phenotyping

(A) Example s-DANNCE predictions (projected 3D poses) for a C57BL/6+BALB/c mouse dyad.

(B) Pairing groups and color codes used throughout the figure.

(C) Inter-animal distance distributions across animals in each group.

(D) Left: map of LLACs (outlined regions) and HLACs (colored regions). Right: LLAC occupancy heatmaps for each animal in each group. LLACs sorted by HLAC and animal center-of-mass (COM) velocity. Below: difference in mean G1 and G2 group LLAC occupancy. G1 and G2 group identity indicated by colored dots as in (B). Scale bar limits, -0.05 to 0.05.

(E) LLAC occupancy PC scores for each animal in each group.

(F) Left: map of LLJCs (outlined regions) and HLJCs (colored regions). HLJC color code as in Figure 4. Right: LLJC occupancy for each animal in each group. LLJCs sorted by HLJC and inter-animal distance. Below: mean-fold-difference LLJC occupancy between groups.

(G) LLJC occupancy PC scores for each animal in each group.

(H) Detailed boxplots for the indicated LLJCs. G1/G2 heatmap dots, $p < 0.05$ (mixed effects), BHFD correction.

Related to Figure S5.

To validate the general, multi-species utility of dyadic embedding, we applied the same framework to two common laboratory mouse strains (Figures 5A–5C and S5). We compared BALB/c to C57BL/6, inbred strains proposed as models of low and high sociability, respectively.^{46,47} Understanding the nature of social behavioral differences in these strains is particularly important for ASD research, as ASD mouse models developed on different backgrounds show pronounced phenotypic differences.^{48,49} When analyzing LLACs, we found behavior in social contexts to be highly consistent across animals of the same strain but observed extensive differences between strains, with C57BL/6 expressing a broader suite of higher velocity behaviors (Figures 5D and 5E). These differences persisted regardless of partner animal background, with LLAC occupancy remaining unchanged between same-strain and mixed-strain pairings. Dyadic embedding, however, revealed phenotypic differences missed when analyzing individual animals (Figures 5F–5H). In mixed-strain pairings, asymmetric engagement LLJCs were upregulated relative to same-strain pairings, with BALB/c mice approaching and inspecting C57BL/6 more often than they would BALB/c partners, in agreement with three-chamber tests showing BALB/c mice prefer social novelty⁵⁰ (Figures 5F and 5H). These results demonstrate how our system enables granular investigation of social behavior across species and genetic backgrounds.

Tactograms capture body-wide patterns of social touch

In addition to behavioral classification, multi-animal pose estimation also facilitates the quantification of social touch, an important facet of social interaction often affected in autism.⁵¹ Despite its clinical and behavioral relevance, social touch is relatively understudied, as doing so traditionally requires subjective and laborious manual annotation. To quantify social touch, we fit a previously developed volumetric model of a rat⁵² to the 3D keypoints of socially interacting animals and automatically identified points of contact between volumetric models (Figures 6A–6C; STAR Methods). Each volumetric model represented the rat body surface as a deformable mesh (6,880 triangle vertices) whose pose and shape were determined by the keypoint-derived rat body structure in each frame. The number and locations of mesh intersections between social partners for a given recording yielded a “tactogram”—the distribution of observed social touches over the entire body over time (Figure 6D). To aid visualization and analysis, we also computed coarser tactogram summaries by binning contact counts over the mesh faces comprising 7 gross body regions (Figure 6D, inset). We validated social contact detection by comparing mesh-derived contacts to three human annotators on 1,100 randomly chosen close interaction frames and found the pipeline achieved human-level accuracy (Figure S6).

We calculated tactograms for each LLJC to examine touch profiles across social motifs. Social touch patterns differed across LLJCs in both the fraction of time spent touching and the patterns of body contact (Figures 6C and 6D). *Mutual interaction* LLJCs displayed a particularly diverse array of touch patterns, from nose-to-nose touching to widespread body contact, potentially conveying different social signals.^{4,53,54} While touch profiles were diverse, they typically fell along two

principal axes capturing the symmetry of contact between social partners (PC1) and the anterior-posterior contact location (PC2) (Figures 6E, S6G, and S6H). Fraction of time spent touching was also variable across LLJCs, ranging from short touches as animals brushed against each other, to prolonged contact when they were investigating (nose to body), allogrooming (head and upper limbs to nape), or huddling (sides of body or heads in contact). *Engagement* and *partner engagement* classes tended to cluster at opposite ends of the partner symmetry axis, with *mutual interaction* in between, serving as an independent validation of our HLJC annotations (cf. Figure 4A).

Tactograms capture dimensions of social interaction not resolved by 3D pose tracking alone, enabling deeper phenotyping in terms of a principal, yet historically understudied, social modality.^{30,55} For instance, amphetamine had a spatially localized effect on touch patterns, with dosed animals displaying a preference for touching the limbs and undersides of undosed social partners (Figures 6F and 6G). These touch patterns could reflect the observed upregulation, in dosed animals, of LLJCs capturing engagement with the front and underside of undosed partners (Figure 4E; Table S1).

Multi-scale embedding reveals patterns of behavior across ASD models

Given that our methods identified amphetamine-induced changes to individual and social behavioral phenotypes, we asked whether they could also delineate how genes affect social behavior by profiling seven genetic rat models of autism: loss-of-function knockouts (KOs) of *Arid1B*, *Chd8*, *Cntnap2*, *Fmr1*, *Grin2B*, *Nrxn1*, and *Scn2a*⁵⁶ (Figure 7A). While these ASD models have been behaviorally phenotyped in mice, their social behavior has either not yet been studied (*Arid1B*, *Chd8*, *Grin2B*, and *Scn2a*) or not yet studied comprehensively (*Cntnap2*,⁵⁷ *Fmr1*,⁵⁸ and *Nrxn1*⁵⁹) in rats, which are far more social animals.^{60,61} Furthermore, reported mouse behavioral phenotypes have been inconsistent and often fail to recapitulate social effects observed in human autism,^{62–64} casting doubt on the utility and validity of mice as ASD models. To probe for social phenotypes in rats and test their validity as models with consistent and relevant social deficits, we recorded and analyzed movies from 142.5 h (25.6 million 3D poses) of lone behavior and 274.9 h (99.0 million total 3D poses across both animals in a dyad) of social interaction across experimental groups (*ARID1B* $n = 4$ heterozygous KOs LE-*Arid1b*^{em1Mcowi}/+, 4 WT littermates; *CHD8* $n = 8$ LE-*Chd8*^{em1Mcowi}/+, 8; *FMR1* $n = 5$ LE-*Fmr1*^{em2Mcowi}/+, 3; *GRIN2B* $n = 3$ LE-*Grin2b*^{em1Mcowi}/+, 3; *NRXN1* $n = 4$ LE-*Nrxn1*^{em1Mcowi}/+, 4; *SCN2A* $n = 3$ LE-*Scn2a*^{em1Mcowi}/+, 3; homozygous KO *CNTNAP* $n = 2$ LE-*Cntnap1*^{em1Mcowi}/*Cntnap1*^{em1Mcowi}, 3). Social behavior was recorded in rat dyads in 30-min sessions, with animals paired within experimental groups in an all-to-all round-robin design (Table S1; STAR Methods).

To map changes in social behavior across rat models of ASD, we quantified and cataloged behavior from tracked 3D kinematics using our multi-scale embedding approach. Additionally, we probed for social touch irregularities, a core symptom of human ASD.^{65,66} Together, these revealed rich animal-autonomous and joint social behavioral repertoires that were largely

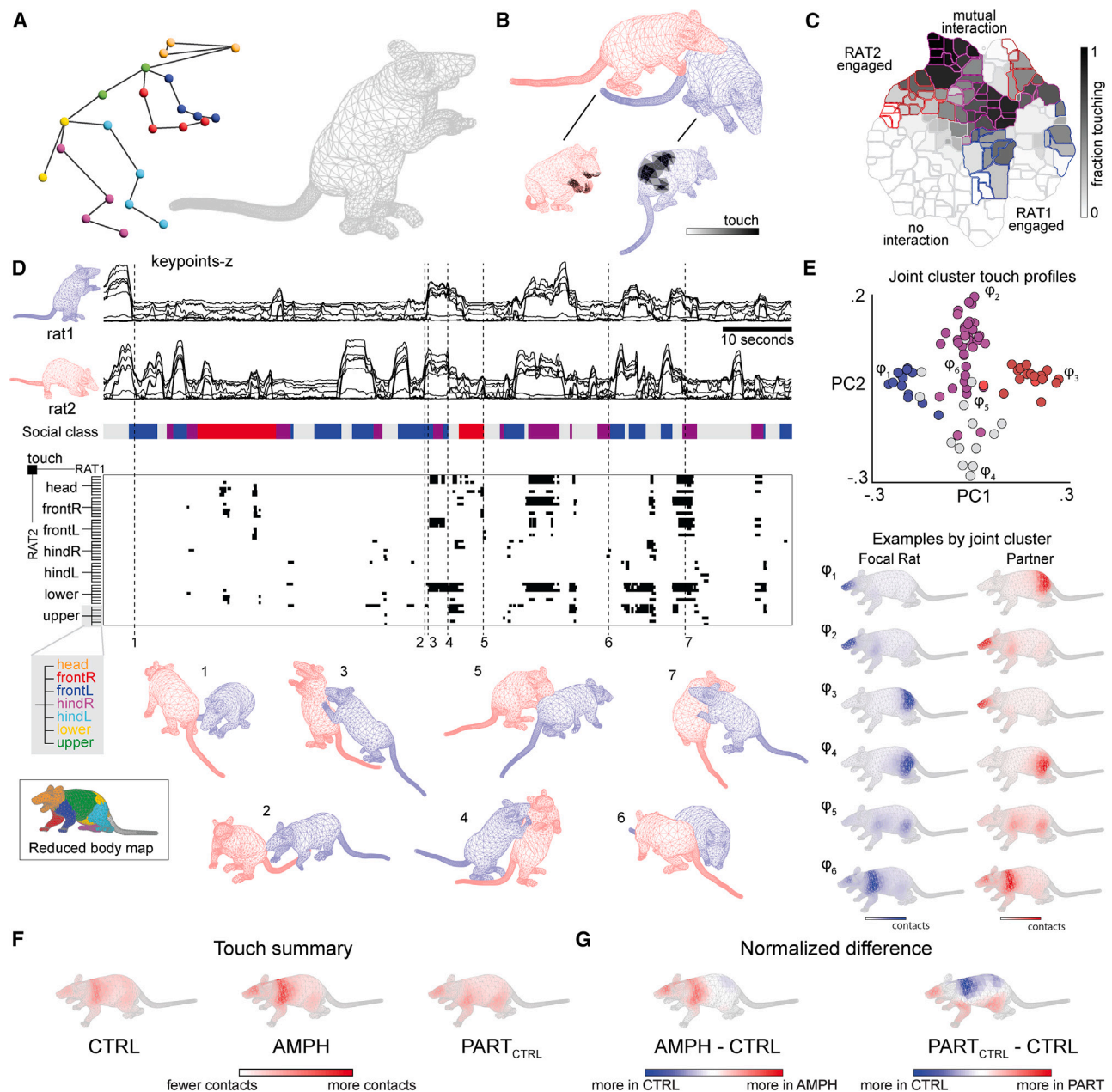


Figure 6. Volumetric body registration for social touch profiling

(A) A mesh model is skinned to 3D keypoints.

(B) Touch detection example for a single frame, and darker fill indicates a larger number of vertex contacts.

(C) Map from Figure 4A, shaded by fraction of time animals touch (any mesh face) while engaged in LLJCs. Outline colors denote HLJC type, as in Figure 4A.

(D) Top: example keypoint traces and corresponding HLJC ("social class," colored as in C), derived from keypoints only. Below: contact events, binned by body region ("Reduced body map"). The raster contains 49 rows, and every 7 is one specific Rat1 body region and the 7 Rat2 regions it could contact. Body meshes are plotted for seven example time points (dashed lines). Colors of example meshes denote animal identity.

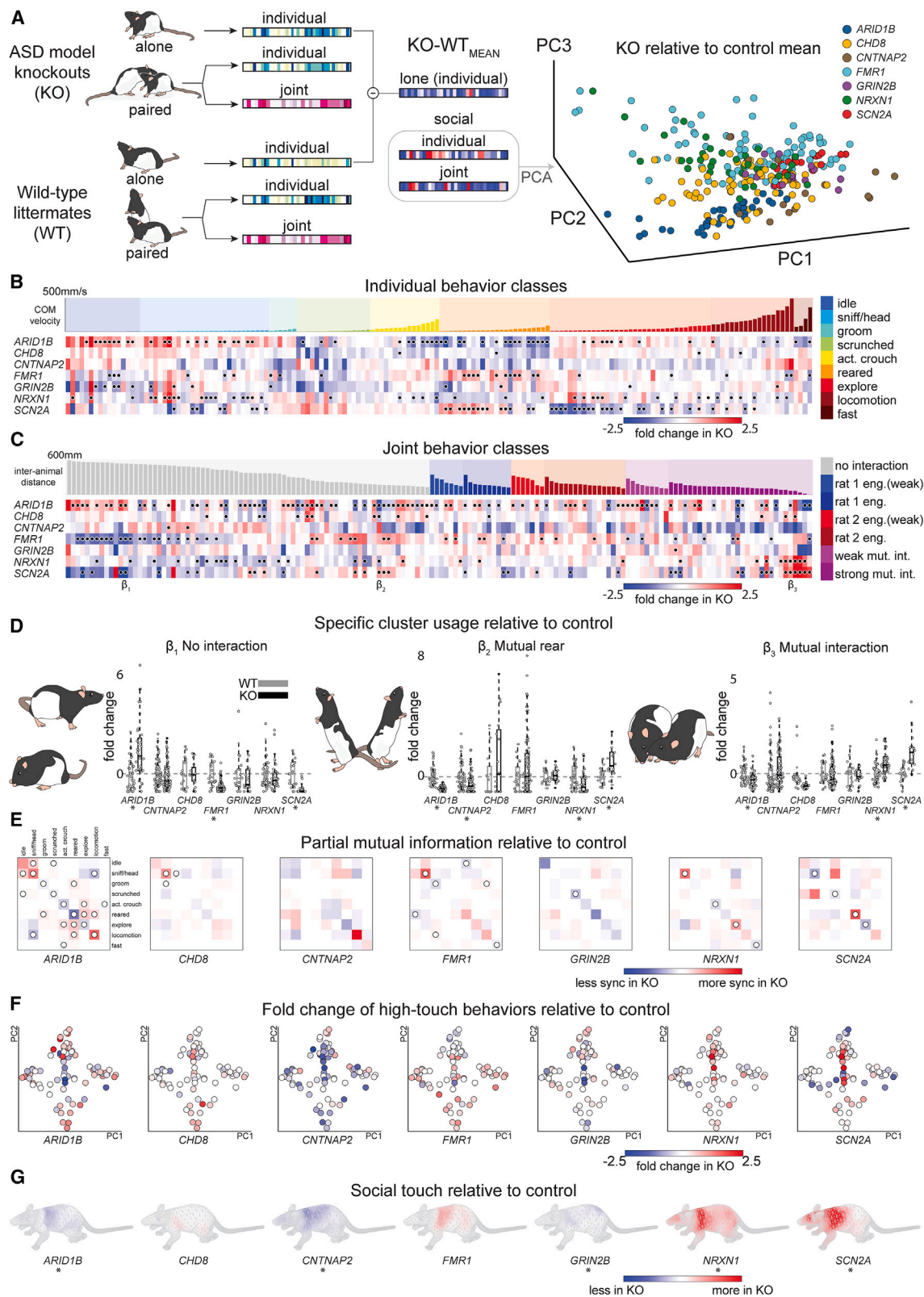
(E) Scatter plot showing PC scores for the mean touch profile of each LLJC (dots, color-coded by HLJC as in C). $n = 396$ paired recordings, 42 animals. LLJCs are derived from keypoints only. An LLJC is only displayed in the scatter if a touch was detected in at least 10% of all frames belonging to the LLJC. Below:

Mean touch profiles, as densities over the mesh, for the indicated LLJCs.

(F) Mean touch profiles for CTRL, AMPH, and CTRL_{PART} animals.

(G) Differences in mean touch profiles between the indicated groups.

Related to Figure S6.



(legend on next page)

consistent within genotypes but distinct across them, with KOs distinguishable from their littermates in their unique LLAC, LLJC, and touch patterns (Figure 7; s-DANNCE repository⁴²). Animal-autonomous interaction profiling of KO animals recapitulated several published mouse phenotypes while revealing social characteristics not resolved by coarse metrics. For instance, while *SCN2A* KO rats exhibited an increase in rearing behaviors, as in mice,⁶⁷ we found that these increases were specific to low-velocity rearing types. Additionally, *SCN2A* KO rats were more likely than WT to perform synchronized rears (Figures 7C–7E). By contrast, the *ARID1B* KOs exhibited decreased rearing, particularly for high-velocity rears, effects not previously reported. Using dyadic embedding, we found that paired animals across all groups engaged in many distinct interactions (e.g., allogrooming, chasing, mutual sniffing, or non-interactive idling), and overlapping but distinct subsets of these interactions were differentially impacted by genotype (Figures 7C and 7D; Table S1). As each social motif is associated with a type of body contact (e.g., nose-to-nose sniffing results in head contact), differences in body-wide touch patterns covaried with the strain-specific expression of social interaction types (Figures 7F and 7G).

The heterogeneity of autism was reflected in rat phenotypes. *ARID1B*, *CNTNAP2*, and *GRIN2B* KOs showed a tendency for reduced contact, driven by a reduction in close interactions. By contrast, *NRXN1* and *SCN2A* KOs were highly interactive and touch-seeking, resulting in increased contact. *FMR1* KOs had a unique presentation, with robust differences from WT in the expression of behaviors where animals were not engaged in interaction. *CHD8* KOs were unique in having particularly high inter-animal variability, a hallmark of autism. These outcomes offer a choice of potential models for studying social deficits, sensory differences, variable penetrance, and non-convergent phenotypes.⁶⁸ Together, our analyses demonstrate an extension of ASD model phenotyping far beyond the status quo and provide a catalog of behaviors across models that can inform mechanistic models of social behavior and help formulate hypotheses about the neural mechanisms that underlie model-dependent differences.

DISCUSSION

We present s-DANNCE, a video-based technology for precise whole-body 3D tracking of interacting animals together with multi-scale analytics for describing structure in social behavior. The platform captures behavior on multiple scales, spanning

detailed, precise kinematics, individual and joint action expression, and body contact patterns, delivering a comprehensive approach for studying social interactions. To illustrate the power of our platform, we profiled interactions in rodents, uncovering effects of drugs and disease not captured by existing techniques, including a spectrum of social changes across seven monogenic rat models of autism.

s-DANNCE accelerates research and opens lines of inquiry

By enabling more objective and sensitive studies of social behavior that scale to capture millions of behavioral events, s-DANNCE can accelerate research into the neural, genetic, developmental, and environmental underpinnings of the social brain and its disorders, as well as screen for novel therapeutics. Because our platform provides automated and quantitative measurements, which we showed are robust across animals, arenas, and sessions, it also lays the foundation for reproducible social behavioral studies across labs. To facilitate reproducibility and large-scale community collaborations, future work will need to explore the question of behavioral re-identification in other labs and explicitly establish standard behavioral definitions and quantitative metrics. To fully capture the complex and multimodal nature of social behavior, s-DANNCE will need to integrate additional measurement modalities, including vocalizations⁷ and relevant physiological variables, such as heart rate and breathing rate. Similarly, to investigate neural underpinnings of social behavior, s-DANNCE must have the capability to synchronize with chronic neural recordings. Relationships between neural, physiological, and behavioral variables could then be probed via recent techniques that map neural activity onto 3D movement features,^{69,70} or, more flexibly, via methods that associate brain and behavioral variables via multimodal fusion.^{71–73}

A better understanding of autism through animal models

Rodent models have delivered key mechanistic and therapeutic insights for many human diseases, but a glaring lack of face validity has muted their impact on autism research. At issue is whether rodent autism models exhibit social deficits, the defining symptom of human autism. In mice, the readouts from social assays have been inconsistent,³ potentially reflecting the stark differences between mouse and human social repertoires. By contrast, rats readily and robustly express many behaviors affected in human ASD, including pro-sociality, cooperation,

Figure 7. Profiling seven rat monogenic ASD models

- (A) Left: illustration of analysis paradigm based on LLACs and HLAC occupancy (“individual”) in lone and social contexts, and LLJC and HLJC occupancy (“joint”). Comparisons are always between animals from the same cohort. Right: PC scores for social LLAC and LLJC KO vs. WT fold change for all recorded pairings (filled circles).
 (B) Heatmaps of mean-fold change from WT littermates in LLAC occupancy for each KO.
 (C) Change in KO LLJC occupancy, visualized as in (B).
 (D) Detailed boxplots for the indicated LLJCs. Fold change from mean WT for each WT recording is also shown (gray box plots). Points denote all subjects across all recordings. Black dots in (B) and (C) and stars in (D) denote $p < 0.05$ (mixed effects) with BH-FDR correction.
 (E) Heatmaps showing differences in mean HLAC PMI between KO+KO and WT+WT pairings. White circles indicate $p < 0.05$ (mixed effects).
 (F) Shifts in LLJCs (as in C) overlaid on LLJC touch PCs (cf. Figure 6E). An LLJC is only displayed in the scatter if animals were touching in at least 10% of all frames belonging to the LLJC.
 (G) Differences in mean touch profiles between KO+KO and WT+WT. * $p < 0.05$ for at least one body part (cf. Figure 6D), $p < 0.05$ (mixed effects). Related to Figure S7.

and age-specific play.^{74,75} High-resolution behavioral assessments will allow us to further probe the utility and face validity of both rat and mouse models. As evidence, we used our precise social phenotyping method to highlight specific social behavioral differences in genetic rat models of ASD and showed that several of the ASD genotypes are associated with strong and consistent deviations from their WT littermates. Importantly, we found evidence of ASD models with both increases and decreases in sociality, as well as genotype-specific differences in social variability across individuals, emphasizing the heterogeneous presentation of these models and the importance of deep phenotyping. In addition, we found divergent body-wide touch patterns across ASD strains, indicating that social deficits can present as both touch avoidance and touch-seeking behaviors, raising questions about whether social motivation can explain autism phenotypes.⁷⁶ Our approach also extended to mouse social behavior, allowing us to identify consistent social differences between strains, including an inbred ASD model (BALB/c)⁷⁷ and capturing shifts in mouse sociality induced by experimental context. Developing these methods further promises a standardized way to probe the landscape of social interactions, how this is shifted in ASD and other neurodevelopmental disorders, and which interventions may reverse these shifts.

Strategies to further enhance 3D social tracking

In the future, s-DANNCE's 3D pose tracking precision and efficiency could be further refined by incorporating recent advances in machine learning and computer vision. For example, s-DANNCE could utilize temporal information, which promotes occlusion robustness in single-animal recordings,^{16,19,35,78} although new temporal modeling strategies with reduced computational costs will likely be needed for multi-animal contexts. Tracking might also be refined by modeling animal pairs jointly, for instance, by sharing network-extracted features between subjects^{79–82} or discouraging keypoint overlap.^{83,84} A more comprehensive, physics-guided approach to tracking, for instance, via the integration of the biomechanical model supporting touch quantification,^{85,86} might also better resolve the most occlusive behaviors, such as rapid tumbling.

Further development of behavioral analyses and feature sets

To explore complex phenotypes with unknown types and degrees of behavioral presentation, we characterized interaction using a large set of 3D kinematic features spanning a wide range of possible behavior-defining movements. However, not all studies necessarily require the same resolution or feature sets. What is best for a specific social analysis will depend on a range of factors, including the behavioral types of interest, dataset size, and experimenter resources. For example, the relative simplicity of traditional approaches that measure a small number of social variables using either manual annotation or sensor-equipped behavioral chambers makes them an appealing choice when tracking a small number of known and pre-prescribed behaviors at coarser resolution. Features derived from fewer keypoints in reduced tracking configurations, such as 2D poses or 3D poses without limbs, could also be used when coarser resolution is sufficient and when social gesturing or contact with limbs is not a

primary research focus. Conversely, studies interested in subtle body language may require keypoint and feature sets expanded from what we present here. While we utilized an unsupervised wavelet-derived feature embedding to flexibly capture behaviors on different timescales, there are several alternative unsupervised behavioral identification approaches that are effective on single-animal pose datasets.^{87,88} Our dataset offers a testbed for adapting these approaches, which might capture different aspects of social behavior, to 3D multi-animal data.

In the future, we envision bolstering our method's flexibility and interpretability via support for a broader range of resolutions and the detection of predefined social behaviors. Predefined behavior detectors and their quantitative feature formulations could be shared with the community and applied to track these specific behaviors in new studies, including in real-time and in closed-loop with behavior or neural manipulations. In parallel, unsupervised analysis will continue to drive the discovery of new phenotypic descriptions using expanded datasets from new experimental contexts, postural tracking resolutions, and measurement modalities.

A foundational repository for high-resolution behavioral analysis

To facilitate continued development of social behavioral quantification and analysis approaches, we are making s-DANNCE available to the community as an open-source Python package.⁸⁹ We have also released our behavioral maps and annotations and our database of 140 million 3D poses, collected from over 500 million images of 6-camera lone and social recordings across 80 animals.⁹⁰ As a large repository of rich, high-resolution social behavior, which includes WT rats and mice, rat amphetamine dosing experiments, and a comprehensive within-litter comparison of multiple rat ASD genotypes, we expect that this resource will foster discovery of characteristics of neurodevelopmental disorders, stimulate new hypotheses of genotype-phenotype mechanisms testable in subsequent experiments, and serve as a reference behavioral atlas to which future quantitative behavioral studies can be compared.

Limitations of the study

As noted above, our study has limitations that should be addressed in future work. There are other core features of social behavior that we are not measuring, for example, facial expressions, gaze, and vocalizations. Capturing these features in freely interacting animals will require further advances in computer vision and, for vocalizations, improved algorithms or instruments for audio source localization. Our current multicamera system is also technically complex compared with conventional social measurement approaches. Reducing the number of cameras, enhancing graphical user interfaces, and systematically probing the relationship between keypoint tracking resolution and behavioral parsing would enhance the adoption of our technology. Although we provided evidence for s-DANNCE generalization to unseen arenas, rat strains, and larger social groups without additional keypoint annotations, there may be contexts where s-DANNCE generalization is weaker. In such cases, we expect s-DANNCE pose tracking would improve given additional annotations for fine-tuning, as we found for mouse and rat

bedding experiments. Relatedly, we did not explicitly quantify reproducibility or test the analysis platform across vastly different social paradigms or setups. We expect the generalizability of behavioral mapping will be facilitated by augmenting pipelines to flexibly support different keypoint configurations and noise profiles. Finally, while we show that our platform extends across species, the study was limited to rodents. A future iteration of the system tested on a larger variety of species will ensure an even broader impact on the study of social behavior.

RESOURCE AVAILABILITY

Lead contact

Further information and requests should be directed to the lead contact, Timothy Dunn (timothy.dunn@duke.edu).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- The s-DANNCE 3D pose tracking package has been deposited on GitHub (<https://github.com/tqxli/sdannce>) and Zenodo (<https://doi.org/10.5281/zenodo.14629231>).
- Code for social behavioral mapping from 3D poses has been deposited on GitHub (<https://github.com/uklibaite/SocialMapper>) and Zenodo (<https://doi.org/10.5281/zenodo.14680563>).
- The social behavior database is available from Harvard Dataverse (<https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7927/H72T-9K94>). The database includes all s-DANNCE 3D keypoint data, individual and social classes (LLACs, HLACs, LLJCs, and HLJCs), and metadata describing the recording (experimental context, experimental group, date of recording, rat ID, amphetamine status, and partner genotype) detailed in Table S1. The database also includes the raw videos and camera calibrations for the entire SCN2A cohort and graphical behavioral data sheets for all rat cohorts.
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

Conceptualization, U.K., T.L., D.A., B.P.Ö., and T.W.D.; methodology, U.K., T.L., D.A., B.P.Ö., and T.W.D.; software, U.K., T.L., D.A., and T.W.D.; validation, U.K., T.L., D.A., and T.W.D.; formal analysis, U.K., T.L., D.A., and T.W.D.; investigation, U.K. and J.F.A.; resources, U.K., T.L., D.A., and T.W.D.; data curation, U.K. and T.L.; writing – original draft, U.K., T.L., D.A., B.P.Ö., and T.W.D.; writing – review & editing, U.K., T.L., D.A., B.P.Ö., and T.W.D.; visualization, U.K., T.L., D.A., and T.W.D.; supervision, B.P.Ö. and T.W.D.; funding acquisition, U.K., B.P.Ö., and T.W.D.

DECLARATION OF INTERESTS

B.P.Ö. is a scientific advisor to dannce.ai. D.A. is an employee of Fauna Robotics. T.W.D. is a co-founder, board member, and paid consultant of dannce.ai.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
Behavioral database	This paper	https://dataverse.harvard.edu/dataverse/socialDANNCE_data
Code for behavioral tracking	This paper	https://github.com/tqxli/sdanne ; Zenodo: https://doi.org/10.5281/zenodo.14629231
Code for behavioral mapping and analysis	This paper	https://github.com/uklibaite/SocialMapper ; Zenodo: https://doi.org/10.5281/zenodo.14680563
Experimental models: Organisms/strains		
Rat: Crl:LE	Charles River Laboratories	Strain code: 006
Rat: LE- <i>Scn2a</i> ^{em1Mcwi}	Medical College of Wisconsin	RRID:RGD_25394530
Rat: LE- <i>Cntnap2</i> ^{em1Mcwi}	Medical College of Wisconsin	RRID:RGD_25330087
Rat: LE- <i>Chd8</i> ^{em1Mcwi}	Medical College of Wisconsin	RRID:RGD_25330088
Rat: LE- <i>Grin2b</i> ^{em1Mcwi}	Medical College of Wisconsin	RRID:RGD_14394515
Rat: LE- <i>Arid1b</i> ^{em1Mcwi}	Medical College of Wisconsin	RRID:RGD_14394518
Rat: LE- <i>Fmr1</i> ^{em2Mcwi}	Medical College of Wisconsin	RRID:RGD_11553873
Rat: LE- <i>Nrxn1</i> ^{em1Mcwi}	Medical College of Wisconsin	RRID:RGD_25330089
Mouse: C57BL/6NCrl	Charles River Laboratories	Strain code: 027
Mouse: BALB/cAnNCrl	Charles River Laboratories	Strain code: 028
Software and algorithms		
Python	Python Software Foundation	python.org
MATLAB	Mathworks	https://www.mathworks.com/
PyTorch	PyTorch Foundation	pytorch.org
Label3D	Aldarondo et al.	https://github.com/diegoaldarondo/Label3D
campy	Severson et al.	https://github.com/ksseverson57/campy
Other		
Cameras	Basler AG	Basler Ace2 Basic USB 3.0
Lenses	Basler AG	Basler 8mm lens
USB3.0 card PCIe	Fresco Logic	Fresco FL1100
Power supply	Corsair Gaming	Corsair RM (2019) 750 W 80+ Gold Certified Fully Modular ATX Power Supply
Memory	G. Skill	G.Skill Ripjaws V Series 32 GB (4 x 16 GB) DDR4-3200 CL16 Memory
Motherboard	Micro-Star International	MSI X299 XPOWER GAMING AC EATX LGA2066
CPU	Intel	Intel Core i9-9900X 3.5 GHz 10-Core Processor
Video cards	Nvidia	PNY Quadro P4000 8 GB, MSI GeForce GTX 1660 SUPER 6 GB VENTUS XS OC
LED arrays	Genaray	Genaray SP-E-500B

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

The care and experimental manipulations of all animals were reviewed and approved by Harvard University Faculty of Arts and Sciences Animal Care and Use Committee. For the control and amphetamine experiments, we obtained 6 male Long-Evans rats (Charles-Rivers, strain 006), aged 9–14 weeks.

For autism model phenotyping experiments, we ordered *Arid1b*, *Chd8*, *Cntnap2*, *Grin2b*, *Fmr1*, *Nrxn1* and *Scn2a* knockouts (only heterozygous except for *Cntnap2* and *Fmr1*) and littermates from the Medical College of Wisconsin (MCW), which produced and maintains these strains in order to facilitate study of rat models of autism (<https://www.sfari.org/resource/rat-models/>). All knockouts are maintained on a Long-Evans background. Animals were ordered in litter- and age-matched cohorts of 6–8 male rats and recorded between 12 and 20 weeks of age. Ear punches were genotyped after birth and experimenters were not blinded to genotype. All knockout animals and littermates were genotyped from tail clippings after all experiments were concluded. Animals arrived housed in groups of 3 or 4 and were allowed to acclimate in home cages for at least 72 hours after arrival. Animals were handled after arrival for weighing and exposure to experimenters. Animals were kept on a normal 12/12 light/dark cycle at a temperature of 22°C and humidity of 30–70% and were housed in ventilated cages with ad libitum food and water. Animals were housed with littermates and isolated at least 48 hours prior to social experiments.

For the mouse experiments, eight C57BL/6 (Strain Code: 027) and eight BALB/c (Strain Code: 028) male mice were obtained from Charles River. The mice were allowed to remain group-housed for the duration of behavioral experiments in cages of four same-strain animals.

METHOD DETAILS

Recording apparatus

All recordings were performed in a custom-built elevated cylindrical arena of 1 meter diameter, where animals could move freely. The circular arena base was water-cut from green HDPE (high-density polyethylene) and the arena wall was constructed from 1mm-thick 60 cm-tall clear polycarbonate sheet to contain animals and prevent escape. The arena was illuminated from above by two white LED arrays (Genaray SP-E-500B, Impact LS-6B stands) and surrounded by a commercial fabric green screen for background consistency and contrast to animals. Six high-speed 2MP Basler Ace-2 Basic cameras equipped with 8mm lenses were synchronized with an Arduino IDE 50 Hz hardware trigger called using campy (<https://github.com/ksseverson57/campy>). A windows PC with 64 GB of RAM equipped with two GPUs (NVIDIA Quadro P4000, NVIDIA GeForce GTX 1660 SUPER, and NVIDIA Quadro RTX 4000 were tested) enabled acquisition from six cameras simultaneously.

Camera calibration

Intrinsic calibration was performed using the built-in matlab Camera Calibration app. Extrinsic calibration was performed with openCV. Calibration parameters were saved in each experiment folder. The entire arena was surrounded with six cameras on tripods which were calibrated prior to experimental recordings. Each day, the current calibration parameters were checked using Label3D (<https://github.com/diegoaldarondo/Label3D>) prior to performing any recordings. In order to check calibration, a short video with a stationary object was captured from all six cameras using campy (<https://github.com/ksseverson57/campy>) and the six views from a single frame were loaded into Label3D. Triangulation from two views was performed for a single point on the object and, if the calibration was still accurate, would correctly identify the chosen point from all remaining views. The calibration was stable enough such that videos could be recorded using the same calibration for weeks, and a new calibration was saved whenever a camera was moved. The pixel resolution is approximately 0.5 mm per pixel on average across camera views.

Behavioral recordings

Experimental protocol details

Before social recordings, all rats were recorded individually in the arena for five consecutive days (except in the case of the *FMR1* cohort, where there were only 3 lone recordings per rat). Animals were then separated into individual housing and kept isolated for at least 48 hours before the start of social recordings. All experiments were completed between 9am and 6pm, however each individual cohort was recorded during the same four-hour time window across days. The arena was cleaned with 70% ethanol each night after recordings were completed and left to dry. Between recordings, a small amount of 70% ethanol was used to wipe the floor and walls of the arena and allowed to dry before the next recording. Rats were randomly paired in an all-to-all round robin format within each experimental cohort. After the complete round robin, additional recordings were performed of rats in random pairings (see Table S1 for a list of each pairing for each rat in every cohort). Animals were all housed and maintained in a facility where they were given food and water ad libitum and monitored by veterinary staff. Cages were cleaned and bedding was replaced each week or as needed. Lights were on from 6am to 6pm.

Lone recordings

Animals were placed directly from the home cage into the acquisition rig for 30 minutes (1800 seconds) on each of five consecutive days with recording starting immediately. The arena was ethanol-wiped and allowed to evaporate dry between recording sessions. All animals from the same cohort were recorded across the same days and each animal was recorded within a consistent 4-hour window across days.

Social recordings

Prior to social recordings, all animals underwent lone recordings in the acquisition rig as described above. Animals were then separated into individual lone cages for at least 48 hours. Animals were recorded in all possible pairs across the given cohort (with their littermates and possibly other animals from the age-matched cohort in cases of small litters). For each social recording, the selected

pair of animals were marked with diluted food coloring in order to simplify center-of-mass (COM) tracking during post processing steps. Animals were placed in the arena in quick succession with allowance for less than 5 seconds of interaction before recording began. Each animal was only recorded in one social session per day except in the case of *CNTNAP2* and *FMR1* cohorts, where multiple recordings could occur in a single day.

Pharmacology recordings (amphetamine)

Prior to lone and social experiments with amphetamine administration, animals were briefly anesthetized using isoflurane, weighed, and injected with 1.25mg/kg amphetamine. After waiting 20 minutes for recovery and acclimation to the injection animals were recorded as previously described. Animals were never dosed with amphetamine or paired with an amphetamine partner in consecutive recordings.

ASD model rat cohorts

The first set of behavioral recordings for each animal was a set of daily 30-minute lone recordings for five consecutive days (3 in the case of the *FMR1* cohort) in the behavioral arena. Each day, animals were removed from the home-cage and placed in the rig by hand after which the experimenter left the room until the recording was complete. Animals were given no prior exposures to the arena for habituation, opting instead to capture the habituation across days. Following the lone recordings, animals were separated into lone housing in home cages and after at least 48 hours social recordings would commence. Each animal was only exposed to one social interaction recording per day (except in the case of the *FMR1* cohort, where multiple recordings could be performed on any given day) and all possible pairings within a cohort were completed before any repeated pairings.

Rat triad recordings

To test the extension of s-DANNCE to >2 animals, we recorded six additional movies of triads from the CHD8 group several days after all lone and dyadic interaction recordings were complete. The three animals were marked using small patches of different colors of food coloring (blue, green, and red) prior to recording and placed into the arena from separate cages immediately prior to the start of recording.

Bedding recordings

To test the extension of s-DANNCE to an arena with bedding, we recorded six additional movies of pairs of Long Evans female rats for 30 minutes each. The animals were marked using small patches of blue and red food coloring prior to recording and placed into the arena, which contained a layer of bedding approximately 1 cm thick.

Mouse recordings

While we developed our method in rats to study their complex social behavior, mice are commonly used in studies of social behavior and as models for ASD symptoms. We compared the lone and social behavior of males from two commonly used lab strains, C57BL/6 and BALB/c, as well as recorded mixed social dyads composed of one mouse from each strain. BALB/c mice are an albino strain often used for immunology research, and have been described as a model of low sociability.^{77,91}

We obtained eight C57BL/6 (Strain Code: 027) and eight BALB/c (Strain Code: 028) male mice from Charles River and allowed mice to remain group-housed for the duration of behavioral experiments. Mice were housed in cages of four same-strain animals. Each mouse underwent two days of lone recording (10 minute experiments) in the behavioral arena after acclimation to the animal facility. Following lone recordings, animals were recorded in pairs for ten minutes each, with each mouse appearing in several recordings in a given day. The white BALB/c animals were marked with a small patch of food coloring on their sides as with the rats to track animal identity. The black C57BL/6 animals required lightening of the fur so half of the animals had small patches of fur on their sides bleached using commercial hair bleach. Bleaching was performed by swabbing a small amount of bleach mixture onto the fur while gently holding the mouse. The bleach was then wiped off and mice were returned to their home cages. This process was repeated across two days for a stronger bleaching effect.

Estimating single-view occlusion

Three sets of social recordings ("SCN2A") and the corresponding s-DANNCE predictions were used for estimating the average occlusion from the top-down and bottom-up view. All 3D points were rotated into the camera coordinate system of the virtual top-down/bottom-up camera and were sorted by their distances to the camera center in descending/ascending order within each separate frame. To account for occlusion by body soft tissue, we made a simplification that a keypoint is considered as occluded if it falls within the occlusion region formed by all keypoints above it in a top-down camera view and vice versa for a bottom-up camera view (Figure S1B). We qualitatively and quantitatively examined the changes in occlusion rates with different occlusion thresholds around each keypoint (5, 10, 15, 20, 25 mm) (Figures S1C–S1E). For all numbers reported in Results, we adopted a threshold of 15 mm.

Animal posture annotation

To acquire 3D posture labels for model training, one human labeler ('Labeler 1', Figure 2F) selectively annotated $n = 423$ unpaired 3D poses from $n = 7$ social recordings of *CNTNAP* rats and annotated one social recording held out from the training set solely for model performance benchmarking purpose ($n = 1373$ frames, both animals, Figures 2I and 2J). To validate the model's generalizability to different rat strains, another social recording of rats with FragileX syndrome was similarly annotated ($n = 1058$ frames, both animals, Figures S3A–S3D). For the inter- and intra-observer error analysis (Figures 2F–2H), $n = 40$ frames with close interaction were evenly drawn from two SCN2A social recordings, where three human labelers annotated both animals in these frames for two rounds with

permuted frame orders. The total number of 3D poses annotated by each human labeler was $n = 160$. All 3D pose annotation was performed using Label3D which triangulates multi-view 2D annotations into 3D. We annotated with a skeleton consisting of 23 body keypoints, as coarsely grouped into 4 body part regions: head (*Snout*, *EarL*, *EarR*, *SpineF*), trunk (*SpineM*, *SpineL*, *TailBase*, *ShoulderL*, *ShoulderR*, *HipL*, *HipR*), forelimbs (*ElbowL*, *WristL*, *HandL*, *ElbowR*, *WristR*, *HandR*) and forelimbs (*KneeL*, *AnkleL*, *FootL*, *KneeR*, *AnkleR*, *FootR*).

Multi-animal 3D centroid localization

We tracked 3D kinematics in interacting animals by first training a multi-instance COM network to roughly locate different individuals and then applying s-DANNCE to infer a keypoint-based skeleton per instance in each frame. Specifically, we construct volumetric representations from multi-camera images, which are anchored at each animal's estimated 3D centroid in the world coordinate system for maximizing the estimation resolution. To determine the animals' center-of-mass (COM) positions, we used a 2D U-Net architecture similar to Dunn et al.¹⁷ We configured and trained different COM networks for lone and social recordings by setting the number of output channels as the number of animals present in the scene. For training, model outputs were optimized against 2D Gaussian-shaped heat maps generated from human annotations of COM positions using a mean-squared Euclidean loss. The 2D COM positions were extracted as the maximally activated position in the 2D heat maps during inference in each camera view and the corresponding 3D COM position was reconstructed as the median across all possible camera pairs.

For each set of experiments, that is, the collection of recordings from a specific cohort of animals, we trained a separate COM network. We labeled approximately 200-500 frames for training each network, making sure to include samples from each animal that appeared in the set of recordings. For social recordings, training frames always included multiple COM labels which were ordered by color (blue, green, red) in order to preserve animal identity. If different sets of colors were used in the same dataset, a separate COM network would be trained for each color pairing.

Multi-animal 3D pose estimation

Graph neural network pose refinement module

For the first stage, we adopted a 3D encoder-decoder architecture to process the 3D volumetric inputs and output the initial estimation of probability distributions for each body marker. For each 3D heatmap associated with a specific keypoint, we performed a differentiable integration trick⁹² to locate the maximally activated location, or center of mass of this 3D volume, as the predicted position of that keypoint. With the initial 3D poses obtained from the regression stage, we defined an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where the nodes were associated with the relative voxel coordinates of body markers $\mathcal{V} = \{v_1, v_2, \dots, v_N\}$ and the edges $\mathcal{E} = \{e_1, e_2, \dots, e_M\}$ linked M pairs of keypoints based on anatomical connectivity of the animal. Accordingly, we defined a $N \times N$ adjacency matrix A where $A_{ij} = 1$ if $e_{ij} \in \mathcal{E}$ else 0. We refer to this graphical representation as a skeletal graph.

We sampled using bilinear interpolation at the maximal activation positions from features maps of the three deconvolution layers. The multi-scale features were concatenated with each keypoint's predicted positions to form each graph node's input features. We then adopted a graph neural network (GNN) to further refine the initial marker localization results. The basic building block of the GNN is a graph convolution module (Figure S2A). Given a generalized graph based input defined as above, each graph convolution block outputs $X^{(l+1)} = \sigma(WX^{(l)} p_l(M \odot A))$ where M acts as an additional weighting parameter that adjusts the contribution of neighboring nodes. The GNN module generates spatial offsets to the initial marker positions in the voxel space. The final marker locations in the world coordinate system were obtained after being scaled by the voxel resolution and translated by the world coordinates of the volume center.

Physical body constraint

We leveraged prior knowledge of rodent body anatomy to constrain the posture solution space. Given all sets of connected markers v_i and v_j , the lower and upper bound of their expected spatial distance $[d^l, d^u]$, we defined the corresponding anatomical plausible range for the lengths of 23 body segments. For better scalability to variably sized animals within and across different rat strains, we reformulated the bone constraints from absolute distances to relative ratios across different segments. In practice, we scaled each segment with the distance between the animal's left ear and right ear landmark. The expected ratios of body parts were pre-computed from ground-truth pose labels and imposed on both male and female individuals as regularization. We imposed body constraints in the form of L2 loss during training.

Semi-supervised learning scheme

The previously described body constraint loss was applied on a mixture of labeled and unlabeled timestamps ($n = 423$ labeled, $n = 910$ unlabeled). To alleviate the tracked marker ambiguity in occluded social scenarios, we further adopted a strong data augmentation scheme that captured the spatial invariability in 3D metric space. During training, we formed each training data batch with augmented copies of one single input volume and imposed batch-wise consistency over all marker location predictions, using a L2 loss $\mathcal{L}_c$.

The overall optimization objective is given by $\mathcal{L} = \mathcal{L}_{\ell_{oc}} + \lambda_b \mathcal{L}_b + \lambda_c \mathcal{L}_c$, where we define $\mathcal{L}_{\ell_{oc}}$ as L1 distance between model keypoint predictions and ground truth 3D positions, if available for the current frames, and λ_b , λ_c are hyperparameters that balance different loss components during training, which are subject to tuning in finding the best-performing combinations.

Model implementation and training procedure

The s-DANNCE framework is implemented in PyTorch. The GitHub repository is made available at <https://github.com/tqxli/sdannce> with codes and detailed instructions for model training, inference and visualization of the tracking results.

For estimating the animals' 2D centroids as the prior step for pose estimation, we trained 2D U-Net COM networks with skip connections. The numbers of channels used in each layer are [32, 32, 64, 64, 128, 128, 256, 256, 512, 512, 256, 256, 128, 128, 64, 64, 32, 32, c = the number of animals present in the scene], where $c = 1, 2$ or 3 . We did not explore more sophisticated tracking strategies as we only focused on animal interactions in dyads and triplets, but anticipate that this step can be conveniently replaced with other existing centroid/identity tracking methods.

For estimating the 3D poses in rats using s-DANNCE, we adopted the same 3D U-Net architecture as Dunn et al. except that we halved the numbers of feature channels in each layer to reduce the model memory footprint. We pretrained the 3D encoder-decoder (i.e., baseline DANNCE) on a subset of the RAT7M motion capture dataset ($n = 88194$) for 20 epochs with an initial learning rate of 0.0001, which was decayed at epoch 10 and 15 by a factor of 10. To compensate for the fine-scale body markers at limb ends missing from the RAT7M dataset, we fine-tuned the 3D encoder-decoder backbone with $n = 1009$ labels with 23 body markers selected from lone rat recordings. The training was done for 450 epochs with a constant learning rate of 0.0001. Note that this adaptation step can be removed without significantly affecting the final performance except for slower convergence. Lastly, we included the GNN module (Figures S2B and S2C) and trained the full s-DANNCE model using $n = 423$ 3D poses collected from social recordings, together with $n = 910$ unlabeled frames for semi-supervised learning as previously described. The model was trained for 100 epochs, with an initial learning rate of 0.0001 decayed at epoch 50, 70 and 80 with a factor of 5, and optimized using the objective function described above, where hyperparameters $\lambda_b = 0.5$ and $\lambda_c = 0.1$. The comparison baseline DANNCE model was trained following the same schedule except for the differences in architecture and optimization objective. All 3D rat pose predictions used for analyses in this paper were obtained from the same s-DANNCE model, including all lone, dyad and triplet recordings.

For generalizing to paired mice and to rats in arenas with bedding, we respectively finetuned the primary rat s-DANNCE model using $n = 111$ 3D mouse pose labels and $n = 211$ 3D rat labels in bedding frames, combining with equal amount of unlabeled samples randomly sampled from frames with inter-animal distances no greater than 120 mm. We did not freeze model parameters during the finetuning and retained the s-DANNCE optimization objective described in section *Semi-supervised learning scheme*. Each model was trained for 40 epochs, with an initial learning rate of 0.0001 decayed at epoch 20 and 35 with a factor of 5.

Comparison with 2D instance-masking method

Han et al.²⁶ described a multi-animal 2D tracking method, which performs 2D pose estimation (DeepLabCut, or DLC) on masked frames from video instance segmentation. To obtain the instance masks required by training and inference of the image masking method, a human labeler annotated $n = 3912$ identity-preserving instance masks in $n = 1980$ frames (severely occluded instances were not annotated), including the entire dyad dataset used for training the baseline DANNCE and s-DANNCE model. These ground-truth (GT) instance masks and corresponding frames were used to train a 2D DLC pose estimation model used for quantitative analyses in Figures S3K–S3M. The DLC model was trained following the default training settings for a ResNet50 backbone for 50 epochs until convergence. The 2D pose predictions yielded by the DLC model were triangulated into 3D after taking the median among all possible camera pairings for each timestamp. For the evaluation, we separately evaluated the image masking method's performance (1) with "GT masks", where the same human annotator annotated animal masks in all test frames, and (2) with "predicted masks", where the training GT masks were used to train a Roboflow 3.0 Instance Segmentation model on the Roboflow platform (starting from the v12 public checkpoint pretrained on MS COCO) and the instance segmentation model was used to annotate the test set.

Individual behavioral mapping

The inferred 3D landmark position trajectories for each recording were used to create a behavioral map which could be used to assign a behavioral label at each time point for each animal across all recordings. We used methods described in several previous works^{36,37} and altered preprocessing, feature selection, and clustering metrics for better performance on the specific data used. Here we briefly describe the method to transform a large dataset of kinematic recordings to behavioral timeseries or ethograms.

All keypoint data were inferred from movies taken at 50 Hz using the same base skeleton (23 keypoints, detailed information saved in 'skeletons/rat23.mat') using s-DANNCE, and preprocessed by median filtering with a 3-frame window followed by a smoothing with a 3-frame window in order to reduce any tracking jitter. First, to normalize for the varying animal sizes across cohorts, the size of each animal was estimated by sampling the distance between two virtual markers (the snout and the tail base) and finding the 97.5th percentile. Inferred points were scaled by this scalar value. Each time point was represented as an all-to-all distance of 23 body marker positions, producing a rotationally and translationally invariant description of the posture of the animal. We performed PCA decomposition on this representation across all recordings to find the top 15 postural principal components which represent the instantaneous posture. To incorporate multi-scale temporal information, we computed the power of each of 25 dyadically spaced frequencies for each of these postural projections. The final representative high-dimensional dataset for each movie incorporated a wavelet decomposition of the PCs (15 modes, 25 frequencies each) as previously described as well as a scaled representation of the height of each joint relative to the arena floor (23 keypoints, scaled by 1/10), and speed for each keypoint (23 keypoints, scaled by 1/2). We found that introducing these features was especially helpful in distinguishing rears (as egocentric alignment

removed raw height data) and small movements. The final high-dimensional representation consisted of a 421-dimensional vector at each timepoint for each individual animal, whether in a lone or social context.

In order to enable behavioral comparisons across all conditions and experimental cohorts, a subset of high-dimensional postural-temporal representations was used to generate a behavioral map which was then used to re-embed each kinematic trajectory. We sampled frames uniformly in time and performed a t-distributed stochastic neighbor embedding (tSNE) for each movie and sampled templates from these initial embeddings to build the final training subspace. The final set of training samples (~45,000 samples) was embedded into a 2D space using tSNE, and each frame from all movies was then embedded into this space to create a density map of behavior across all experiments. Applying a watershed transformation to this space ($\sigma = 1$) resulted in 163 spatial clusters along this 2D density which represent fine-grained behavioral clusters which we refer to as LLACs (low-level action clusters). These clusters were further grouped into nine coarse categories which we refer to as HLACs or high level action clusters (*idle*, *sniff*, *groom*, *scrunched*, *reared*, *active crouch*, *explore*, *locomotion*, and *fast*). The fine- and coarse-grained cluster labels were applied to behavioral recordings taken from both lone and paired animals.

Behavioral synchrony calculation

Synchrony analysis was performed by considering the HLACs assigned simultaneously to paired animals across each recording as described in Klibaite and Shaevitz.¹¹ To identify co-occurrence patterns of HLACs in interacting animals, we calculated a 9x9 synchrony matrix (S) for each paired time-series using the partial mutual information equation:

$$S_{ij} = p(\text{HLAC}_1 = i, \text{HLAC}_2 = j) \log \left(\frac{p(\text{HLAC}_1 = i, \text{HLAC}_2 = j)}{p(\text{HLAC}_1 = i) p(\text{HLAC}_2 = j)} \right)$$

where HLAC1 and HLAC2 refer to the fractional occupancies of each of the nine HLACs (high-level action clusters) for animal 1 and animal 2. Each entry in S, in units of bits, captures how likely the pair of behaviors are to occur together above chance. The underlying behavioral distributions were calculated independently for each recording. We report the average for S across experimental groups (Figure 3F; Extended behavioral data sheets⁴²).

Social behavioral mapping

We extended the previously described methods for unsupervised individual behavioral mapping to produce behavioral clusters capturing the postures of two interacting animals, where features capture the relative posture and dynamics. Several changes were necessary to perform multi-animal social embedding and capture social motifs. First, since the identity of an animal matters within a social context, each social cluster is from the perspective of one of the animals. In practice, this means that features are calculated twice for each interaction recording, with the ordering of keypoints flipped for capturing the perspective of the second rat. We used fewer principal components (6) to describe the shared posture and positioning of interacting animals than in the single animal case, as the emphasis was in capturing stereotyped instances of shared movement instead of a combinatorial description of the movements of each animal. The time scale used for the joint wavelet decomposition was also slower in order to accommodate slower and less precisely coupled movements than in the case of single-animal limb coordination. Shared and individual-animal features were combined to produce the final high-dimensional representation. This combination of features capture aspects of the posture and movement of each animal and emphasizes the shared measures such as distance and orientation between animals as well as the dynamics of shared movement.

To perform the joint embedding, we first calculated the all-to-all distances between selected 3D keypoints (*snout*, *spineF*, *spineM*, *spineL*, *tailBase*, *shoulderL*, *shoulderR*, *handL*, *shoulderR*, *handR*, *hipL*, *footL*, *hipR*, *footR*) on each animal and performed online-PCA to determine six basis vectors to capture the shared postural representation across all movies. Wavelet decomposition at dyadically spaced frequencies ranging from 0.2-5Hz was performed on the resulting projections obtained from representation for each social recording. Additional features including distances between specific body parts and angles of heading deflection between animals in 3D were derived from the inferred keypoints and added to the high-dimensional time-series describing the shared dynamics.

In order to sample across all contexts and balance the final embedding in consideration for rare motifs, templates were retrieved from each social recording and pooled to form a training set for embedding using t-SNE, as described for the animal-autonomous mapping. The resulting training set and 2D t-SNE map were then used to re-embed each recording, resulting in a final representation that was clustered using the watershed algorithm. The corresponding embedded values were used to assign each frame a cluster number. These social clusters capture possible animal configurations incorporating not only static features such as distance and orientation, but also properties of these shared features across a span of time scales. Each of the resulting 156 clusters (five clusters could be eliminated because of a lack of samples) was assigned a low-level description based on viewing randomly selected snippets of frames which were assigned the given label (Video S4). The movies contained both animals but were described from the perspective of the focal animal, giving rise to categories that aim to capture how engaged the focal rat and partner are, leading to a description of how engaged each animal is and how symmetric this engagement appears to the human eye. While many clusters (76 of 156 total) were classified as *no interaction*, these clusters still contain valuable information about the interaction between animals and range from low-level descriptions like “*laying on opposite sides of the arena with little to no activity*” to “*both performing active rears while less than one animal-length apart*”. While there does not appear to be active inter-animal engagement in case of the latter description, this cluster captures a synchronized movement.

Importantly, each animal within a social pairing is assigned its own social cluster label at every frame, as the social embedding retains the identity and perspective of the given animal. For example, one animal may correspond to a cluster where, from that animal's perspective, it is "chasing" the other animal, whereas the partner will instead correspond to a cluster that captures "being chased". These 'perspectives' can be resolved to a single interaction-level annotation based on an agreement matrix derived from the human-annotated descriptions. This intermediate description can be used to evaluate how much one-sided or mutual interaction occurs throughout a social recording and how interaction progresses over time (Video S5).

Manual annotation of clusters

Two human annotators independently viewed movies of reconstructed 3D skeletons randomly sampled from each behavioral cluster from individual and joint mapping. Sampled clips were picked at random from all experimental movies, from time points where the given label was assigned for at least 25 subsequent frames or 500ms of consecutive kinematics. For individual (animal-autonomous) clustering, each cluster was assigned a short description and a coarse category (Table S1; Video S3). For joint clustering, each cluster was assigned a description and the interaction was given a coarse social category (*no interaction*, *lightly engaged*, *engaged*, *partner lightly engaged*, *light mutual interaction*, and *mutual interaction*) intended to capture the level of asymmetry and intensity of the sample movies (Table S1; Video S3). Any clusters where annotators were in disagreement were reviewed together until a consensus was reached to produce the final set of coarse behavioral cluster descriptions presented here.

Estimating social touch with mesh models

To estimate points of contact between socially interacting rats, we registered a biomechanical model paired with a deformable mesh model to s-DANNCE keypoints and quantified intersections between meshes. Here, we will describe the biomechanical model and accompanying deformable mesh model, the registration process, and the algorithms used to determine contacts.

Biomechanical model

We used a biomechanical model paired with a deformable mesh model as a tool to estimate the volumes of interacting animals at each frame by registering the biomechanical model to the keypoints inferred with s-DANNCE. The biomechanical model of the rat was developed as part of a previous study⁵² to resemble the bone length and body mass distribution of Long Evans rats in the laboratory and work with the MuJoCo physics simulator. The pose of the biomechanical model has 74 degrees of freedom (DoF). These include the global Cartesian position of the model's root (3 DoF), the quaternion defining the orientation of the root (4 DoF), and a set of transformations denoting the joint angles of each body part relative to its parent in an acyclic tree that denotes the relationships between connected body parts (67 DoF).

To account for differences in animal sizes, we isometrically scaled the biomechanical model according to the segment lengths estimated with s-DANNCE. For every session, we calculated the median segment length for each animal and each session, seg_{as} , and computed the median segment length across all sessions, seg_{med} . The scaling factor for a particular animal and session, s_{as} , was then defined as the ratio $\frac{seg_{as}}{seg_{med}}$.

Deformable Mesh

The biomechanical model described above includes a deformable mesh model that was created to approximate the shape of a rat. The deformable mesh model is implemented using the MuJoCo skin asset. Briefly, the deformable mesh is composed of a set of 6880 vertices connected to form a set of triangular 10752 faces that together approximate the shape of a rat. The position of each vertex and the orientation of its normal is influenced by the positions of one or more bones in the biomechanical model. Each bone contributes to the position of a set of vertices with a weight that specifies how much the position and orientation of the bone contribute to the position of the vertices. The 3D position of each vertex is calculated by weighting the influence of its corresponding bones. Normal vectors for the mesh are computed from the 3D vertex positions and the faces derived from connections of vertices.

To better approximate the volume of animals of different sizes, we specify an inflation parameter for each animal which expands or contracts the deformable mesh along the normal vectors by a given distance. Similar to the scaling of the biomechanical model, we used the distribution of bone segment lengths estimated from s-DANNCE to determine the inflation parameter. The inflation parameter for a particular animal and session, i_{as} , was calculated as follows

$$i_{as} = .003m * \frac{seg_{as}}{seg_{med}}$$

The value of .003 m was determined through visual inspection by overlaying the deformable mesh models of different inflation parameters atop the multicamera images of the animal with the median segment length. The inflation parameter allowed us to scale the size of the deformable mesh model to account for variability in the animal sizes, or growth that occurred throughout the collection of the dataset.

Registration

Our biomechanical model was designed to approximate the general skeletal structure of the rat, and not skin, fat, muscle or cartilaginous tissue. This presented a problem when trying to register the biomechanical model to s-DANNCE keypoints, as several keypoints correspond to bodily landmarks that were not present in our biomechanical model. To address this problem and register the biomechanical model to the s-DANNCE keypoints, we used a custom implementation of simultaneous tracking and calibration

(STAC).⁹³ STAC iteratively optimizes two quantities in an alternating fashion to calculate the biomechanical model poses that best explain the keypoint positions: a set of learned offsets that relate points on the biomechanical model to s-DANNCE keypoints (m-phase) and the joint angles of the biomechanical model (q-phase) (Figures S6A–S6E).

In the m-phase, STAC optimizes a set of offsets that each relate a reference point on the biomechanical model to an s-DANNCE keypoint. Specifically, the m-phase uses L-BFGS-B to minimize the mean squared error between s-DANNCE keypoints and ‘fictive keypoints’ defined by applying the set of offsets to the set of reference points. The reference point for a given s-DANNCE keypoint was defined as the location of the parent of the body closest to the keypoint. The offset was defined as a 3D vector extending from the reference point in the reference frame of the parent body. This optimization occurs over a small training dataset that will be described in greater detail below. At each step of the optimization, we adjust the set of offsets by an epsilon step, compute the new fictive keypoints using MuJoCo’s forward kinematics, and compute the empirical gradient to determine the direction of the offset update.

In the q-phase, STAC optimizes the joint angles (expressed as a set of quaternions) via least squares optimization to minimize the mean squared error between the s-DANNCE keypoints and the fictive keypoints. At each step of the optimization, we adjusted the joint angles by an epsilon step, computed the new fictive keypoints using MuJoCo’s forward kinematics, and computed the empirical gradient to determine the direction of joint angle updates. Importantly, the offsets were fixed during the q-phase, enabling the optimization of joint angles for a given set of offsets and frames. We applied this procedure to each frame serially, and initialized the pose of the biomechanical model using the pose from the previous contiguous frame. In the first frame, the biomechanical model’s base pose was used to initialize the optimization.

Applying STAC involves two steps: calibrating the biomechanical model to s-DANNCE keypoints, and registering the calibrated biomechanical model to new data. To calibrate the biomechanical model for a given animal and session, we created a training dataset from the initial 10 seconds of data (500 frames) and optimized the offsets and joint angles through three alternating iterations of the q- and m-phases. For each q-phase, we optimized each frame in the 500-frame trajectory, while for each m-phase, we randomly selected 50 frames from the 500 frame training set. To register the calibrated biomechanical model to all of the data from a given session, we reused the offsets from the calibration step and performed a single q-phase optimization over all frames.

Contact determination

To estimate the points of contact between two animals, we automatically identified frames in which the mesh models for two interacting animals intersected. For every frame, we posed the biomechanical models representing both animals in MuJoCo according to the poses registered with STAC and logged the positions of the vertices comprising the mesh models. To detect intersections between meshes, we constructed a bounded volume hierarchy from the vertices and faces comprising one animal’s mesh and queried whether any vertices comprising the other animal’s mesh were contained within the volume. When a vertex was found to be inside of the volume, we logged the identity of the penetrating vertex, the identity of the face closest to the penetrating vertex, the identity of the vertex on that face closest to the point that was the shortest distance between the face and the penetrating vertex, and the frame number. For the purposes of analyses, contacts were defined by these pairs of vertices and the frame number.

In general, the procedure described above is asymmetric such that the contacts estimated by constructing a bounded volume hierarchy from one animal of a pair does not equal the contacts estimated by constructing a bounded volume hierarchy from the other animal. We symmetrized it by performing the procedure twice, swapping the identity of the animal used to construct the bounded volume hierarchy, and combining the collection contacts resulting from both iterations. When applied over a recording, this process generates a collection of vertex-vertex pairs encoding the spatial location of contacts on both animals and the times at which they occurred.

Contact validation

To validate the performance of our automatic contact estimation method, we compared its predictions to those of human annotators on a set of frames in which two rats were in close proximity. Using a collection of six 30-minute sessions sampled evenly across the genetic models of ASD, we randomly sampled 1100 timepoints in which the centers of mass of both animals were less than 150 mm apart. We then used Label3D to view the animals from all six camera angles simultaneously. We defined seven regions on the rat’s body to consider in our contact validation (Head, Left forelimb, Left hindlimb, Right forelimb, Right hindlimb, Upper body, and Lower body). The boundaries between each of the body parts was defined by grouping bones of the biomechanical model into each of the seven categories, noting the bones with the highest weight for each vertex, and assigning each vertex to one of the seven categories according to the category of its maximum-weight bone. For each frame in the dataset, three human labelers manually labeled contacts between each of the body part pairs from which we estimated the interlabeler variability in manual annotation. To estimate the intralabeler variability, we had two labelers repeat their annotations for a subset of 500 timepoints.

To validate automated contact estimates, we applied the contact estimation method to the timepoints in the manually labeled dataset. For each timepoint, we organized the resulting contacts into the seven body part categories, denoting a body part contact pair if there existed a vertex contact pair with vertices corresponding to each body part. Because contact events are relatively sparse, we used balanced accuracy as a metric to quantify the performance of our contact estimation method, equally weighting the accuracy of touches and non-touches.

We considered three baselines against which to compare the deformable mesh model’s performance: a ‘keypoints’ baseline in which we derived a contact-detection model directly from s-DANNCE keypoints, the interlabeler variability, and the intralabeler variability. For the keypoints baseline, we aimed to estimate how well one could estimate social contacts by using only the pairwise distance between keypoints on social partners. We defined a contact as any time in which two keypoints were within a threshold of

30 mm apart, and assigned each keypoint to one of the seven coarse regions to compare to manually labeled contacts. We estimated the optimal threshold value with a line search. For the interlabeler variability, we compared the contact estimates across the three individuals, computing the average balanced accuracy across all labeler pairs. For the intralabeler variability, we compared the contact estimates from individuals on one day to those from the same individuals on the same data, labeled on a different day, and computed the average balanced accuracy across individuals.

Touch PCA

To produce a low-dimensional representation of touch within each low-level joint cluster (LLJC), we isolated frames when animals were in each LLJC and found the fraction of time each of the body zones (head, limbs, upper front, and upper rear) were in contact with the other animal for the focal rat and the partner rat (Figure 6). We concatenated these profiles into one 14-dimensional vector and normalized this representation to produce a touch profile for each LLJC where touch occurred for at least 10% of total frames (84 of 156 LLJCs). We performed PCA on the set of normalized touch profiles and found that the first two principal components captured over 75% of the variance (Figures S6G and S6H). We used this 2-dimensional space as a complementary way to visualize behavioral changes in ASD knockout models (Figure 7).

QUANTIFICATION AND STATISTICAL ANALYSIS

After reducing each behavioral experiment to a ‘behavioral profile’ or probability vector that captures the occupancy of each of the fine-grained behavioral clusters (only Action Clusters in the case of lone recordings, both Action and Joint Clusters in the case of social dyads), we compared these profiles between experimental groups. Because there are multiple observations from the same individual when they are paired with different partner subjects within an experimental group, we fit a hierarchical mixed model to perform statistical comparisons.^{94,95} This prevented a false assumption of independence between samples from the same individual. Specifically, we fit a linear mixed effects model (LME, Matlab function *fitlme()*) to observations (e.g., the usage of a specific cluster in each recording session) given a fixed effect for experimental group (e.g., ASD KO vs. WT littermate) and a random effect for intercept grouped by subject identity.

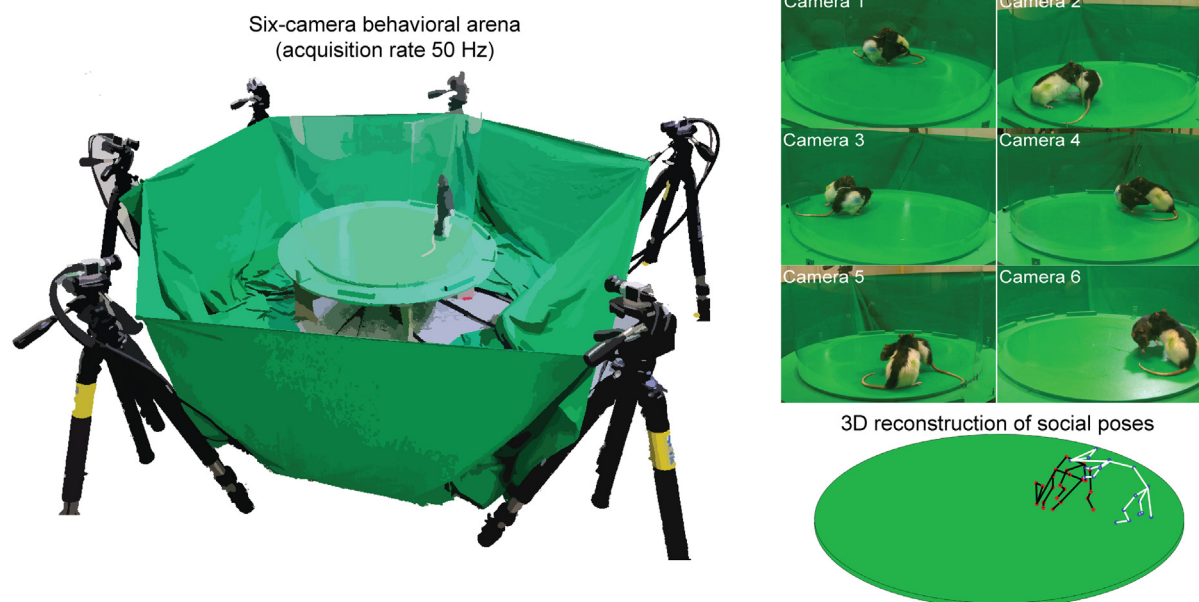
We performed statistical comparisons at two levels: one at the level of the behavioral profile for each group and another at the level of occupancy for each low-level behavioral cluster. In the first case, we compared a behavioral profile from two groups (for example an ASD knockout model and the corresponding WT littermates) by calculating the Jensen-Shannon Divergence (JSD) for profiles measured in each session to the average from one of the groups (Figure S5; Table S1). These JSD values were the observations used to fit the LME model accounting for subject identity, as described above.⁹⁴ At the level of individual clusters between groups, for each cluster we performed the same LME fit, with cluster occupancies as observations, and corrected the resulting p-values for multiple comparisons using the Benjamini-Hochberg False Discovery Rate (BHFD) method (MATLAB function *mafdr(..., 'BHFD', 'true')*).

We reported the p-value and JSD between the mean of the two groups for each set of comparisons as well as the number of behavioral clusters from each coarse group (HLAC and HLJC) that was up- or down-regulated in the cohort of interest (Table S1; Figure S7). We wrote Matlab wrapper functions *testGroupLme* and *findSigBeh* to identify the group and cluster level significance, respectively, and have included them in the SocialMapper code repository.

We also used the same LME process for determining statistical significance for PMI (Figure 7E) and overall changes in social touch (Figure 7G) as we did for individual cluster comparisons. Except instead of cluster occupancies being the observations, we used the values in the flattened synchrony matrices (for PMI) and the fraction of touches occurring in each of the 7 reduced body parts (for overall changes in social touch).

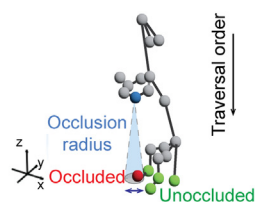
Supplemental figures

A



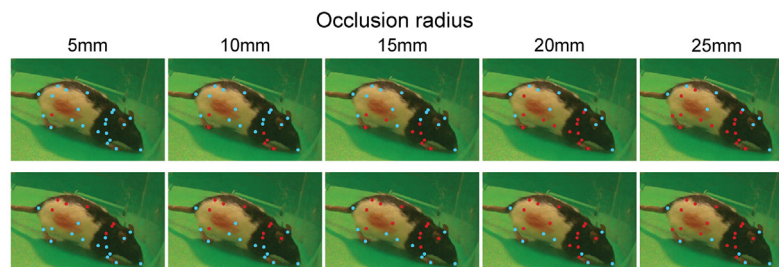
B

Top-down view
occlusion estimation



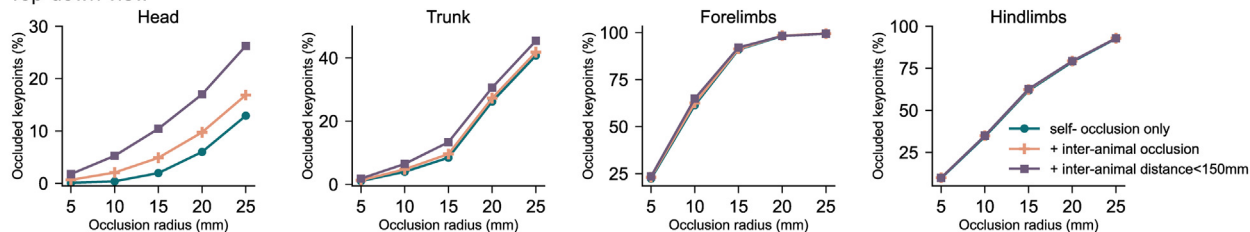
C

Top-down view
Bottom-up view



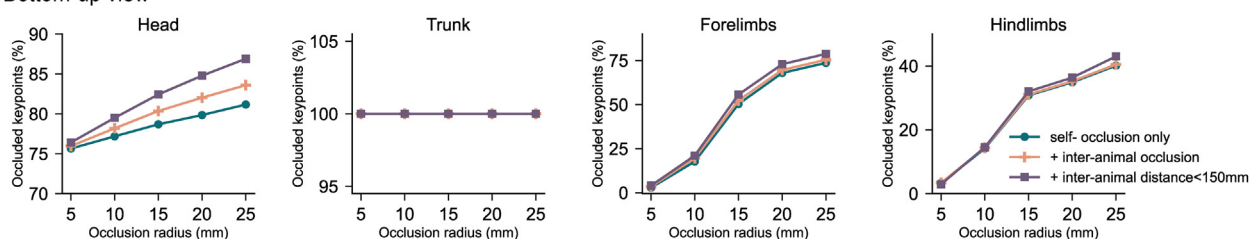
D

Top-down view



E

Bottom-up view



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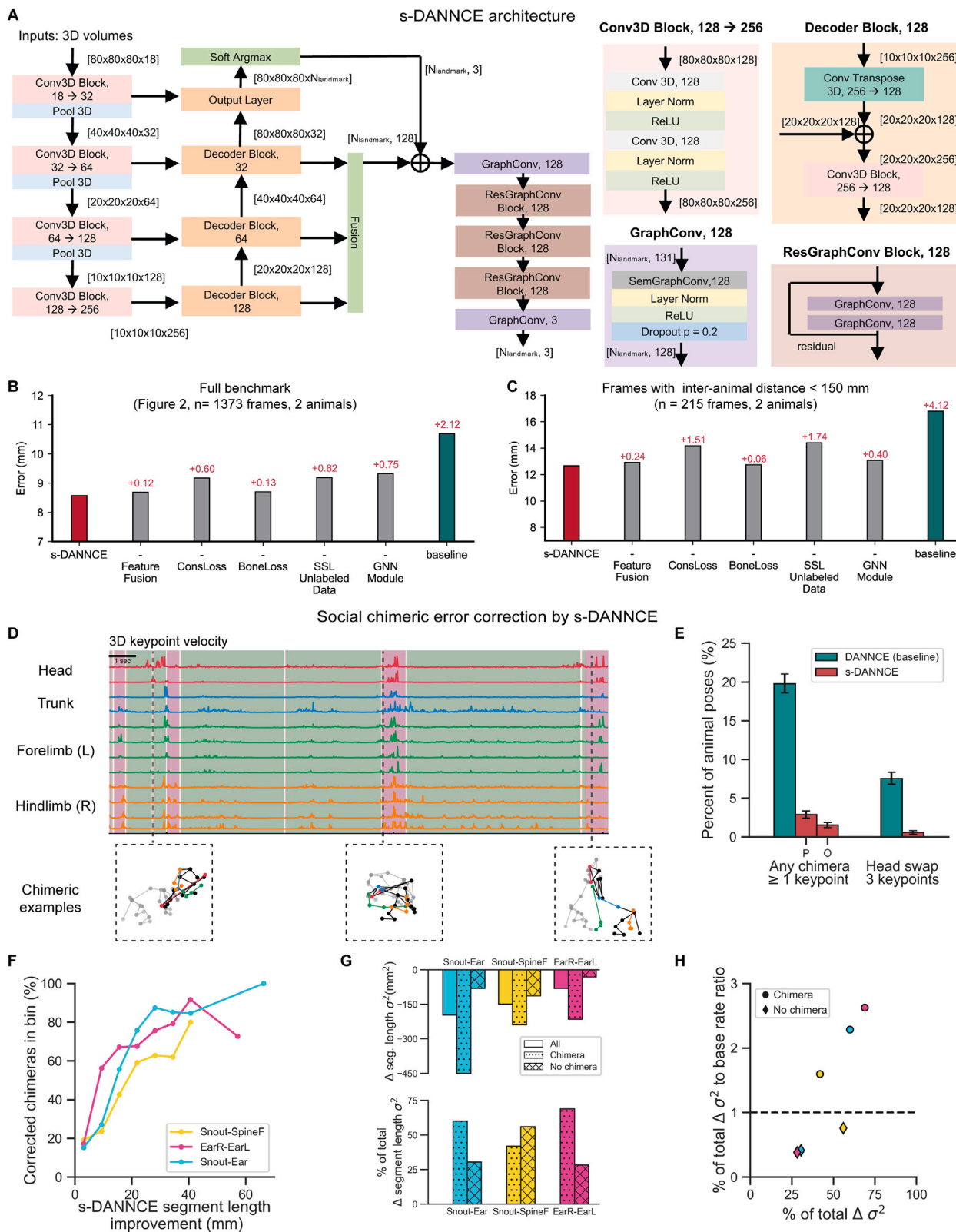
Figure S1. Multicamera behavioral arena and occlusion analysis in social animals, related to Figure 1

(A) Multicamera behavioral arena setup. Left: a circular behavioral arena is surrounded by six synchronized 50 Hz color cameras for recording rat interactions. Top right: Single time point capture of six-camera acquisition from the behavioral arena. Bottom right: the corresponding 3D social poses inferred by s-DANNCE from frames.

(B) Schematic demonstrating how occlusion between keypoints is estimated in an imaginary top-down camera view. Blue color indicates the keypoint of interest and its affected region following the traversal order. Similar procedure was applied for estimating bottom-up view occlusion.

(C) Qualitative examples of how occlusion varies with different occlusion radii. Red indicates that the keypoint is assumed to be occluded.

(C and D) Quantitative comparisons of the proportions of occluded keypoints in the animals' head, trunk, forelimbs, and hindlimbs, respectively, from the top-down and bottom-up view. The occlusion rates are independently computed for different occlusion radii (5, 10, 15, 20, and 25 mm), and except for self-occlusion from the animal's own body (blue), we optionally consider occlusion from social interactions (orange, purple).



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Figure S2. s-DANNCE corrects chimeric errors for improved anatomical stability, related to Figure 2

(A) Detailed schematic of the s-DANNCE neural network architecture.

(B and C) Ablation studies of different modeling components in s-DANNCE. The (–) symbol refers to the ablation of that specific component from the best-performing s-DANNCE model. For the complete descriptions of each modeling component, refer to [STAR Methods](#).

(D) Failure cases of baseline DANNCE in interacting animals. Top: first-order velocity traces of 12 selected body markers in one animal engaging in social interactions ($t = 20$ s). Background colors separate out the dynamics during close inter-animal distances (pink: < 150 mm). Bottom: qualitative examples sampled from timestamps with sudden velocity spikes. Strong correlations between close inter-animal distances, implying social activities, and chimeric errors in DANNCE predictions are observed.

(E) Percentage of poses with chimeric errors in close interaction frames (inter-animal distance 150 mm or less), for both baseline DANNCE and s-DANNCE. The overall s-DANNCE chimera rate includes partially fixed chimeras (P; chimeras with fewer keypoints than baseline DANNCE) and all others (O). Error bars are 95% confidence intervals.

(F) Plot showing the rate of s-DANNCE chimeric error correction as a function of s-DANNCE segment length improvement magnitude (the amount of reduced segment length deviation from the mean). Larger s-DANNCE segment length improvements were predominantly from frames where s-DANNCE corrected chimeras, illustrating the impact of chimeric error correction on segment length stability metrics.

(G) Top: change in segment length variance (σ^2) from baseline DANNCE to s-DANNCE for the indicated body segments, shown for all frames ($n = 6,500$) and frames with ($n = 1,706$) and without ($n = 4,794$) baseline DANNCE chimeras. s-DANNCE variance reduction (improved stability) was larger in frames with baseline DANNCE chimeric errors. Bottom: variance reduction in frames with and without baseline DANNCE chimeras, expressed as a percentage of the variance reduction in all frames.

(H) Scatter plot showing the base rate-relative share of segment length variance reduction for frames with and without baseline DANNCE chimeras. Variance reduction in chimera frames has a disproportionately large effect on overall segment length stability.

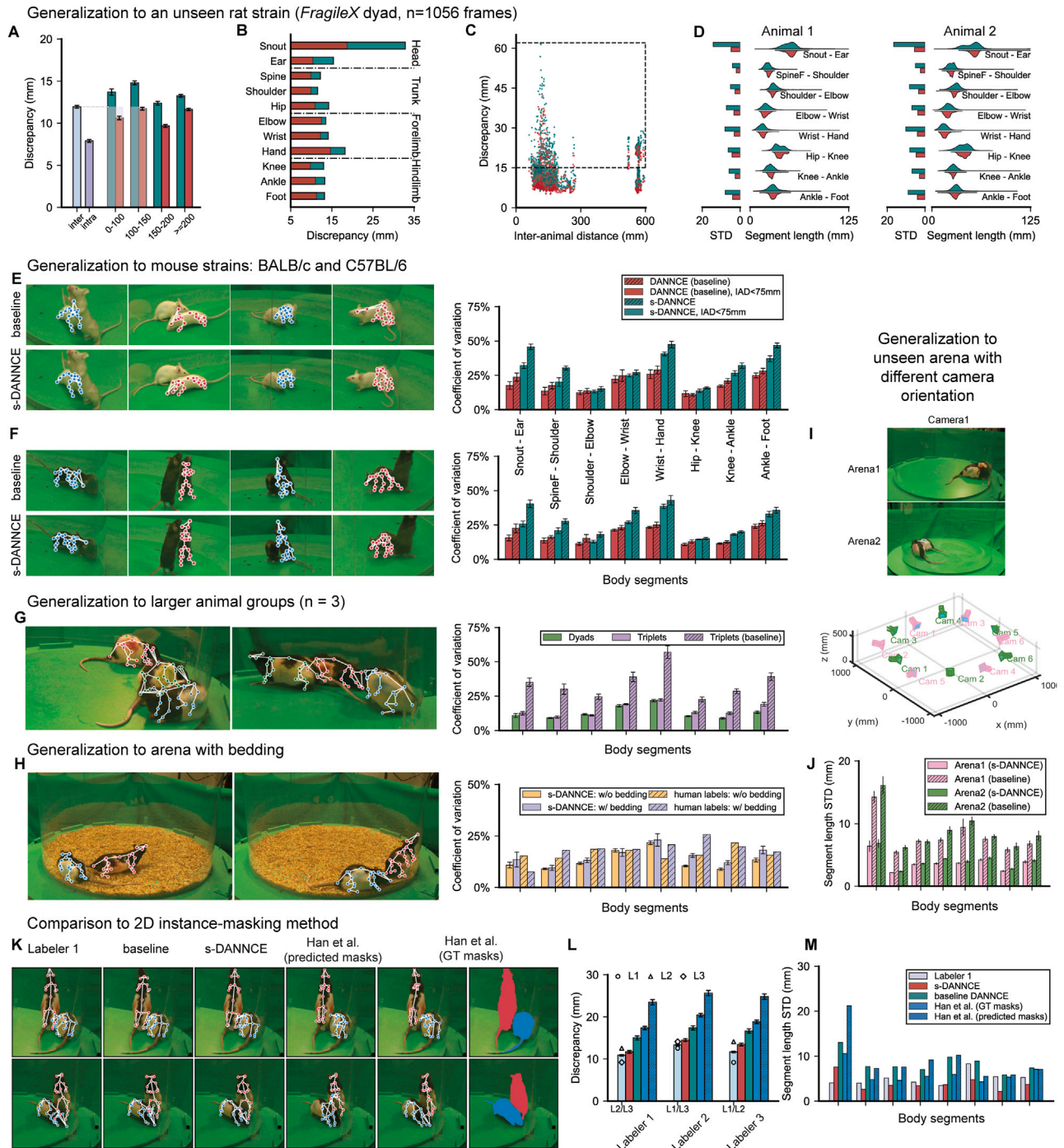


Figure S3. s-DANNCE tracks social behavior across strains, setups, species, and experimental conditions, related to Figure 2

(A) Bar plots of landmark localization discrepancy over a different recording of FMR1 KO (*fragileX*) rats ($n = 1,056$ frames, 2 animals). Frames are grouped by the inter-animal distances (0–100, 100–150, 150–200, and ≥ 200 mm). Error bars are 95% CI at keypoints. The average intra- and interdiscrepancies of the three human labelers (light blue and purple bars, respectively) are computed from Figures 2F and 2G.

(B) Mean errors with respect to different body landmark positions ($n = 504$ frames within 150 mm inter-animal distance).

(C) Scatter plot of landmark localization discrepancy as a function of inter-animal distance.

(D) Distributions of body segment lengths as derived from different model predictions in each animal in the recording. Symmetric body segments are merged for conciseness.

(E and F) Generalization to mice. Using the pretrained rat models as initializations, we, respectively, trained baseline DANNCE and s-DANNCE models using $n = 111$ additional 3D pose labels collected in paired mice. Left: qualitative tracking performance comparison between baseline DANNCE and s-DANNCE in BALB/c

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mice. Right: coefficient of variation (%) in segment lengths derived from different model predictions in $n = 6$ dyad recordings, shown for all frames ($n = 180,000$) and for frames with inter-animal distance ("IAD") no greater than 75 mm ($n = 39,476$). Same analyses were repeated for C57BL/6 mice in (F). Out of a total of $n = 180,000$ frames, $n = 27,052$ frames have inter-animal distance no greater than 75 mm. (E and F) error bars are 95% CIs across animals.

(G) Generalization to larger-sized rat groups. Left: qualitative tracking examples from a triplet of Long Evans rats when directly applying the s-DANNCE rat model. Right: tracking stability in terms of coefficient of variation (%) in segment lengths, as the number of simultaneously tracked animals in the arena scales ($n = 2, 3$). For the $n = 4$ triplet recordings (each with $n = 90,000$ frames), we sampled $n = 6$ dyad and $n = 12$ lone recordings of SCN2A knockouts (KO), each with a total of $n = 12$ instances for comparison. Error bars are 95% CIs across animals.

(H) Generalization to an arena with bedding. All predictions were made by a model fine-tuned from the pretrained rat model using $n = 211$ additional 3D pose labels collected in an arena with bedding. Left: qualitative tracking examples of female Long Evans rats within an arena with additional bedding. Right: coefficient of variation (%) of segment lengths derived from model predictions and human annotations, with and without bedding in the arena. For model performance, we compared $n = 6$ dyad recordings of Long Evans female rats in the bedding arena with $n = 6$ dyad recordings of male SCN2A KOs. For human performance, we compared the $n = 211$ pose labels used for training the bedding model to the $n = 160$ 3D poses annotated by labeler 1 as used in [Figures 2F and 2H](#). Error bars are 95% CIs across animals.

(I and J) Generalization to an arena with different camera orientation.

(I) Visualization of two arenas with different camera orientations. Pose labels collected in arena 1 were used to train the rat models, and we examined the shifts in tracking performance when directly generalizing the model to the unseen arena 2.

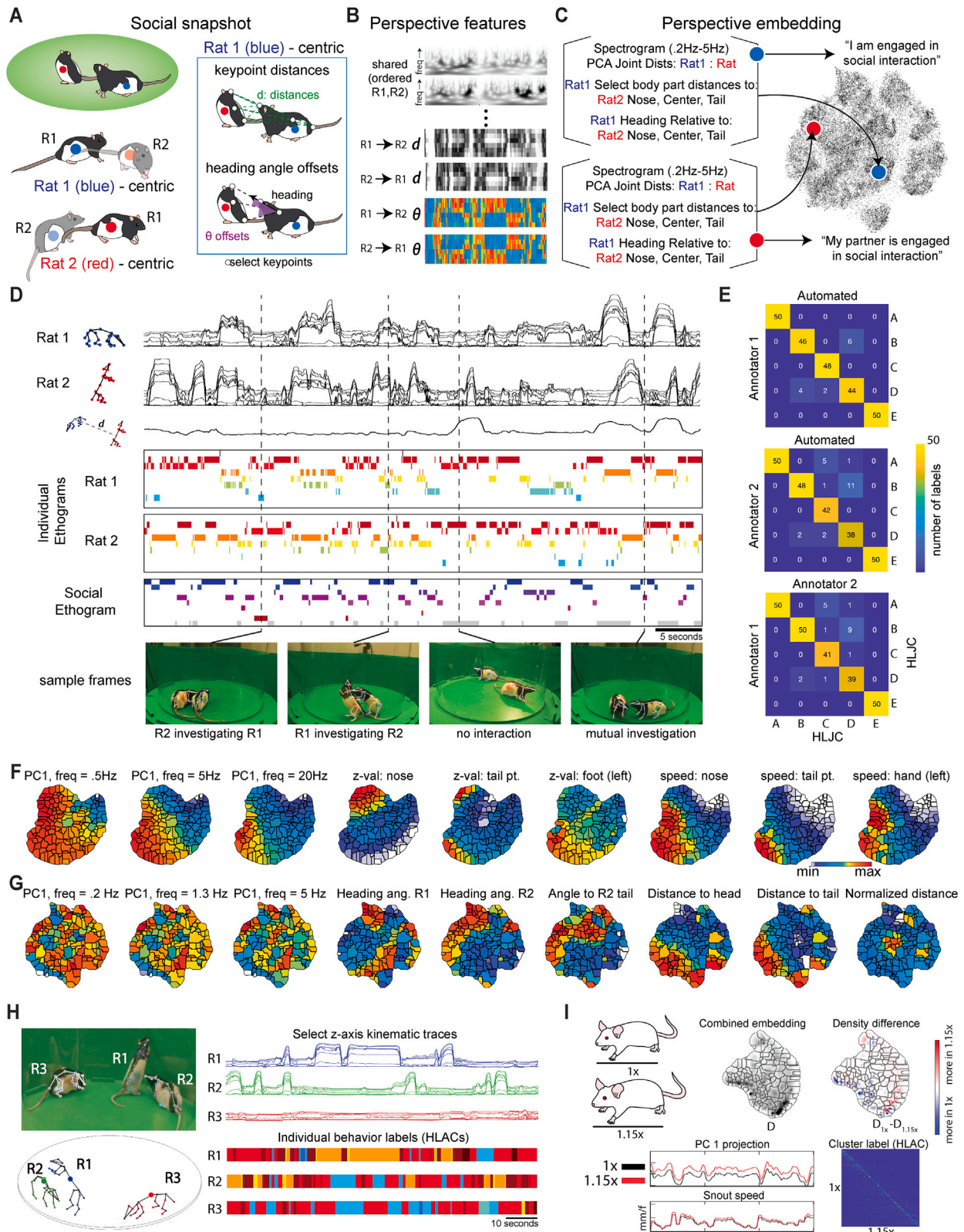
(J) SD in segment length derived from predictions of s-DANNCE and baseline DANNCE, respectively, in each arena. $n = 6$ social recordings of male Long Evans SCN2A KOs were sampled to represent tracking performance in arena 1, and $n = 6$ social recordings of male Long Evans with amphetamine injections were sampled for arena 2. Frames were evenly balanced by the corresponding inter-animal distances with a bin size of 100 mm to accommodate for shifts in behavioral profiles. In contrast to baseline DANNCE, the s-DANNCE model demonstrated better robustness against shifts in camera positioning. Error bars are 95% CIs across animals.

(K–M) Comparison to a multi-animal tracking method,²⁶ assessing 3D pose estimation performance of 2D keypoint triangulation when animals are first segmented out of images with 2D instance masks.

(K) Qualitative comparison between human ground truth pose labels ("labeler 1"), baseline DANNCE, s-DANNCE, and instance-masking methods. Ground truth (GT) instance masks were never used for baseline DANNCE and s-DANNCE. For the 2D instance masks, we report results from using predicted masks and from using GT masks. For GT masks, a human labeler annotated $n = 3,912$ identity-preserving instance masks in $n = 1,980$ dyad images, inclusive of the entire dyad dataset used for training s-DANNCE. These GT masks and frames were used for training the 2D pose estimation model (DLC) used for all subsequent evaluations. The labeler also annotated GT masks in all test frames, which were used for the "GT masks" evaluation. For the "predicted masks" benchmark, the training GT masks were used to train a Roboflow 3.0 Instance Segmentation model on the Roboflow platform (starting from the v12 public checkpoint pretrained on MS COCO). This model was then used to predict masks on the test set. For the 2D pose predictions yielded by the instance-masking method, we adopted a triangulation protocol that took the median of 3D reconstruction among all possible camera pairings.

(L) Bar plots of inter-labeling discrepancies between human and model annotations. Human, baseline DANNCE, and s-DANNCE data are reproduced here from [Figure 2G](#). Error bars are 95% CIs across keypoints.

(M) SDs in segment length. Human, baseline DANNCE, and s-DANNCE data are reproduced here from [Figure 2H](#).



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Figure S4. Description and quantification of dyadic feature embedding, related to Figure 4

(A) Visualization of the dyadic (joint) embedding. Each point in the social recording contains two animals with identities maintained. From the perspective of a single (focal) subject animal, the shared postural and dynamic information is embedded into a joint map to parse joint behavior motifs.

(B) The embedding from the perspective of each animal is processed individually, and the order of shared kinematic features maintains the “perspective” for each individual.

(C) Each point in behavioral space corresponds to a single-animal perspective of a social interaction. Behavioral labels then describe the shared interaction from the perspective of that animal.

(D) Sample kinematic traces, distance between two animals, low-level action clusters for each individual, low-level joint clusters from the perspective of the first rat, and sample frames with overlaid skeletons from inferred s-DANCE are shown for a minute-long bout of interaction.

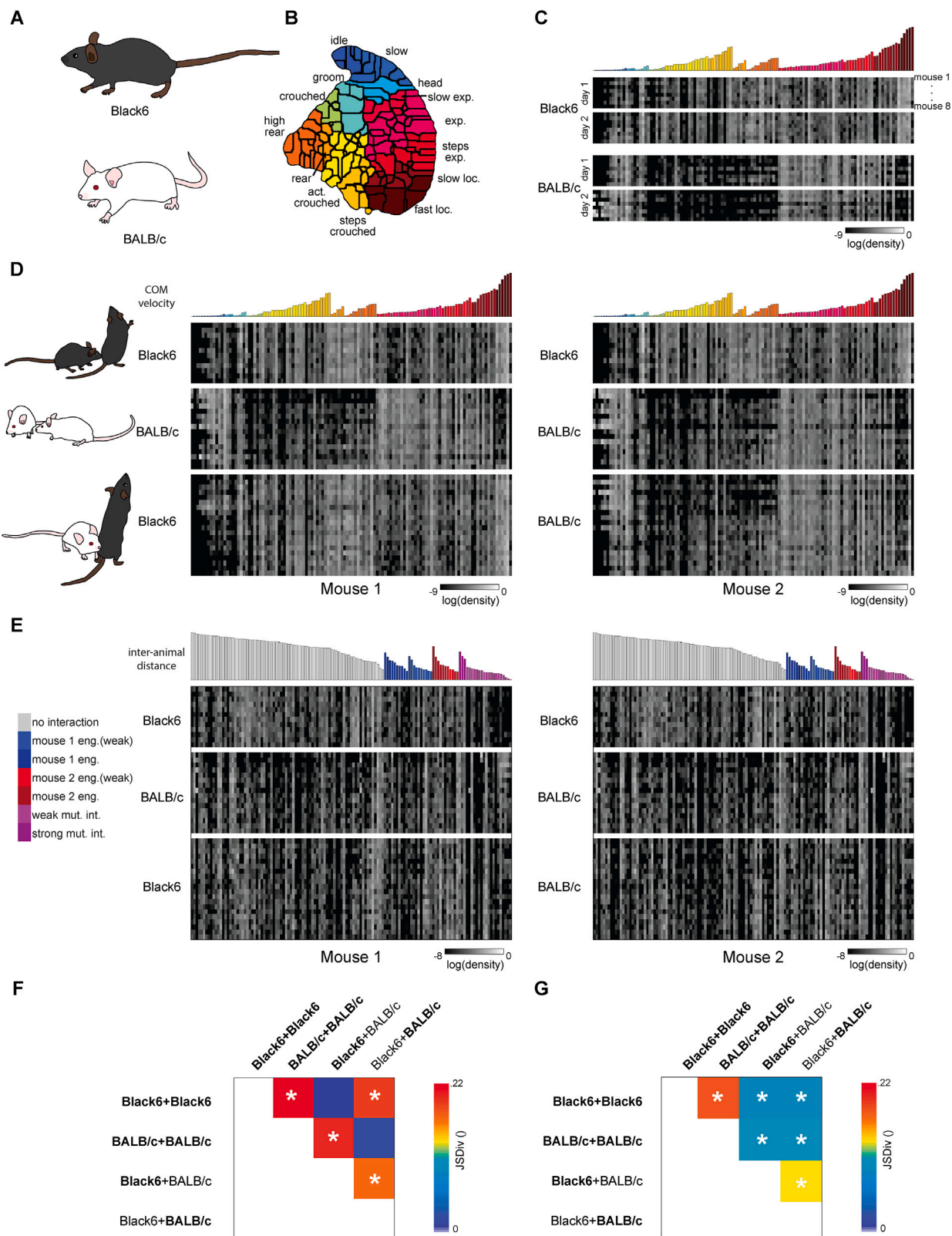
(E) To assess the accuracy of dyadic embedding clusters, we compared manual annotation of 250 snippets of social data consisting of 50 snippets each from five low-level joint clusters derived from our unsupervised social clustering. Two annotators were given instructions to assign one of five behavioral groups to each cluster: *no interaction* (far apart, little movement), *rat1 engaged* (rat1 following rat2, rat2 avoiding), *rat2 engaged* (rat 2 investigating rat1 from behind), *mutual interaction* (close head-to-head, allogrooming), and *no interaction* (mutual rearing). The annotators had 95.2% and 91.2% agreement with the original cluster labels, and there was 92.0% agreement between the labelers.

(F) Mean values of sample features used for action clustering.

(G) Mean values of sample features used for joint clustering.

(H) Behavioral analysis for rat triad recordings shown as a sample frame (top left), snapshot of sDANCE-inferred skeletons (bottom left), z-traces of select keypoints over 1 min of recordings (top right), and ethograms representing high-level action clusters (HLACs) for the corresponding behaviors (bottom right).

(I) An embedding was constructed by combining all lone mouse recordings as well as a copy with the distance between keypoints scaled up by 15% to test the effect of un-normalized body scale differences on behavioral mapping. A single behavior map was produced using this dataset, and the total density (top middle) and difference between the unscaled and scaled densities (top right) are shown. Example traces of three different features are plotted (bottom left), showing how they are affected by body scale. A matrix shows the discrepancy between samples in the 1.15× dataset that are assigned to the same clusters as original 1× samples (bottom right).



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Figure S5. Mouse social phenotypes across trials, related to Figure 5

(A–C) Results from lone C57BL/6 (Black6) and BALB/c sessions.

(B) Mouse lone behavior map.

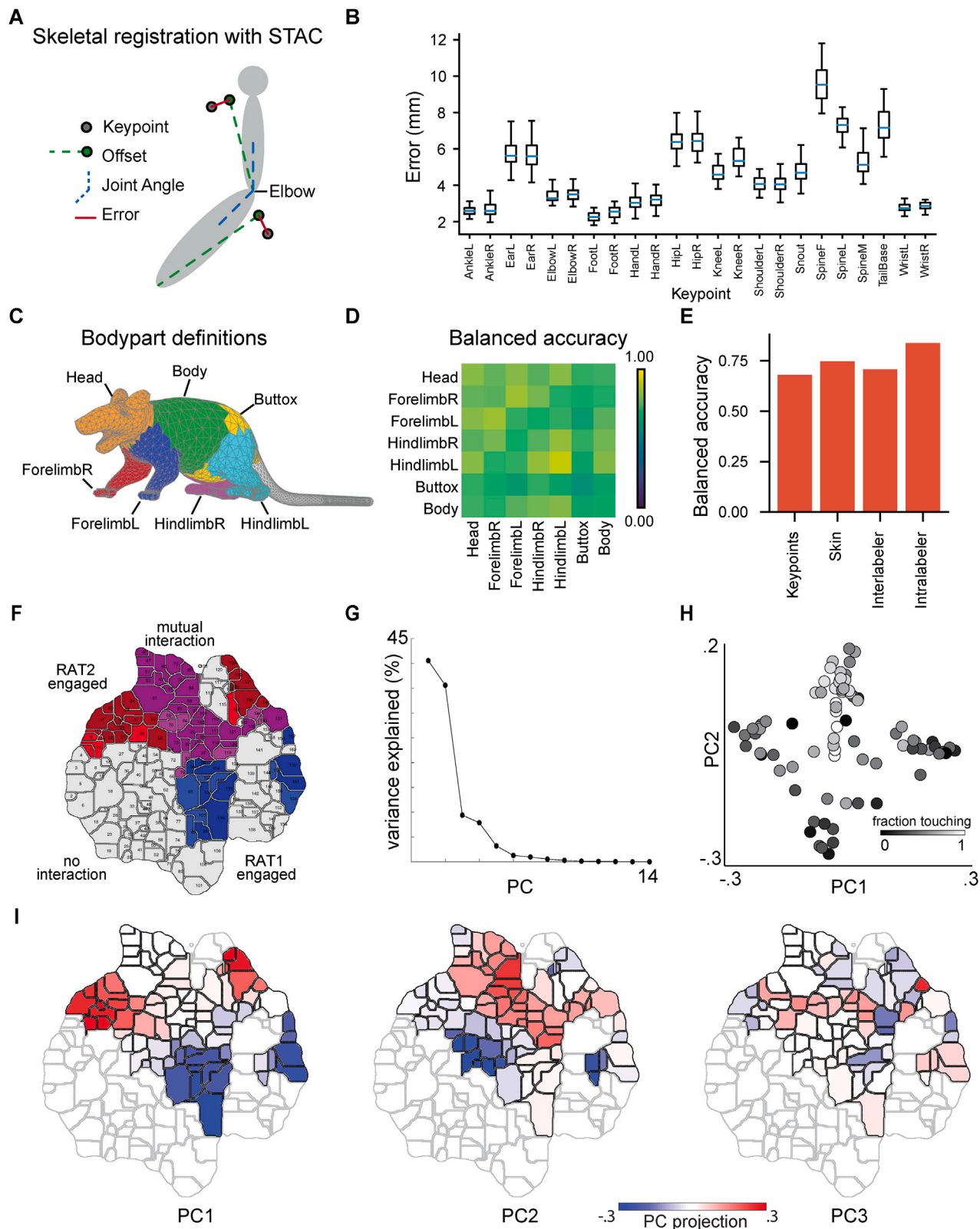
(C) Heatmaps LLAC occupancies for lone sessions (columns, sorted by HLAC and COM velocity) for all animals, across two different days of recording.

(D) Heatmaps of mouse social LLAC occupancies across all recording sessions for the indicated strain pairings, for both animals in the pairing (“mouse 1,” left; “mouse 2,” right). LLACs (columns) are sorted by HLAC and COM velocity.

(E) Heatmaps of LLJC occupancies for the same recordings as in (D). LLJCs are sorted by inter-animal distance and HLJC type (legend on left).

(F) Table of Jensen-Shannon divergence (JSD) between the mean (across animals in the bolded strain) LLAC occupancies for different pairing groups (see also [STAR Methods](#)). * $p < 0.05$ (mixed effects).

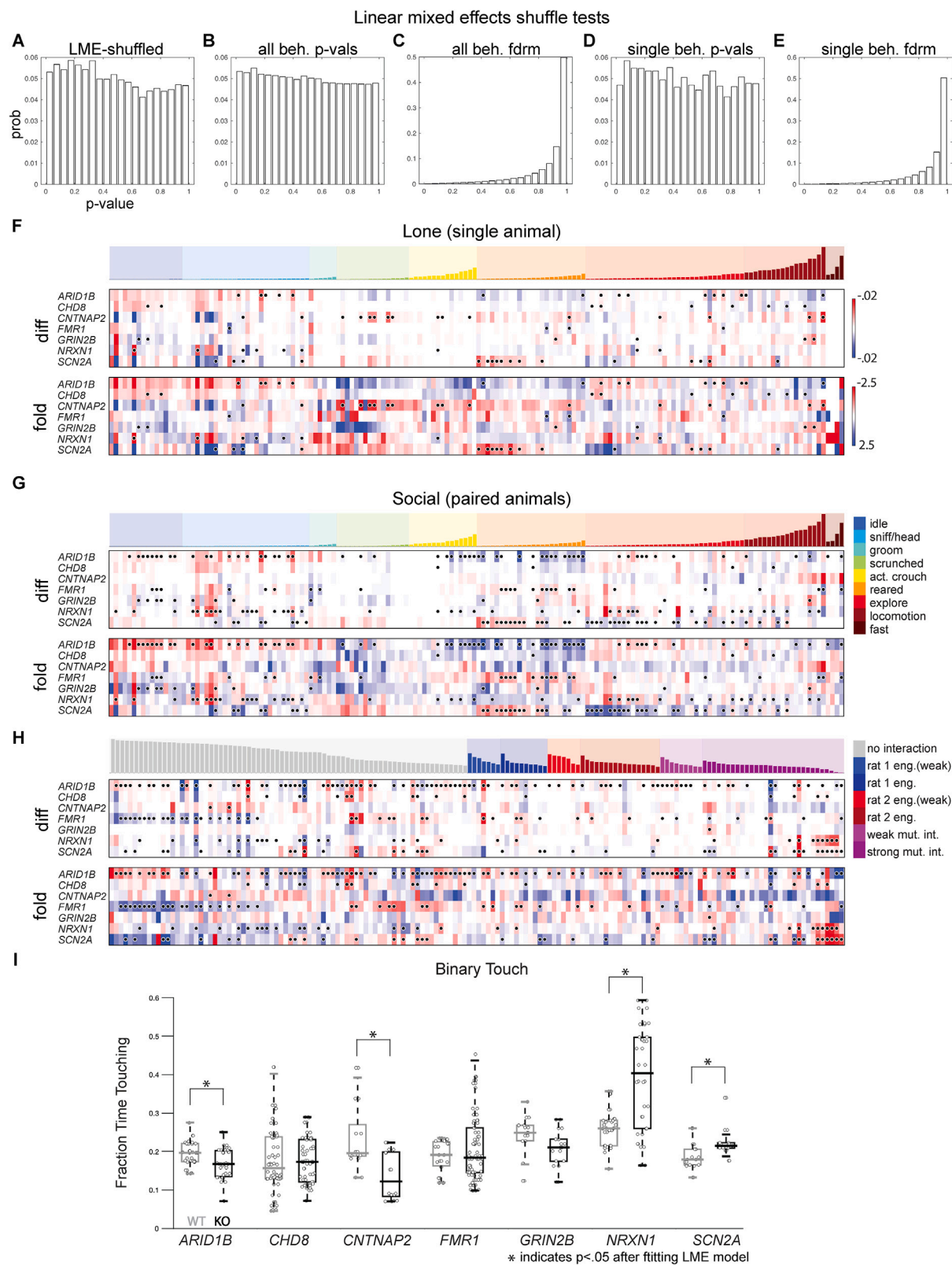
(G) Table of JSD between the mean (across animals in the bolded strain) LLJC occupancies for different pairing groups (see also [STAR Methods](#)). * $p < 0.05$ (mixed effects).



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Figure S6. Social touch estimation and profile analysis, related to Figure 6

- (A) Schematic of skeletal registration. simultaneous tracking and calibration (STAC) iteratively optimizes a set of learned offsets that relate points on a skeletal model to keypoints and the joint angles of the skeletal model that best explain the keypoint positions.
- (B) The distribution of average registration errors for all keypoints across sessions. Blue lines indicate the median, box limits indicate the 25th and 75th percentiles, and whiskers indicate the maximal point up to 1.5 times the interquartile range from the nearest box limits.
- (C) Schematic depicting the definitions of body parts used for manual contact labeling.
- (D) Balanced accuracy of contacts estimated with the deformable mesh relative to manual contact labeling for all body part pairs.
- (E) Average balanced accuracy of contact quantification methods and human labelers. Balanced accuracy is the average of the true positive rate (the fraction of contacts that were correctly identified) and the true negative rate (the fraction of non-contacts that were correctly identified). The keypoints method defined contacts as any time in which two keypoints on social partners were within 30 mm of one another (see [STAR Methods](#)). Interlabeler balanced accuracy was the average accuracy between all pairs of distinct human labelers. Intralabeler accuracy was the average balanced accuracy of human labelers labeling the same data on two different days. The skin model achieves a balanced accuracy greater than both the keypoints approach and the accuracy between labelers.
- (F) Joint clustering with high-level category labels.
- (G) Variance explained per PC on 14-dimensional (coarse/reduced body map) normalized touch densities for animal1/animal2.
- (H) Fraction of time touching for likely touch (touching $\geq 10\%$ of the time) clusters on PC axes (refer to [Figure 6](#)).
- (I) Loadings onto the first 3 PC axes for social clusters (non-touch clusters outlined in gray).



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Figure S7. Summary of comparisons for ASD model rat cohorts, related to Figure 7

- (A) Linear mixed effects model (LME) model fits were performed on condition- and identity-shuffled data from the WT and amphetamine social experiments. 10,000 trials were run with different permutations of group assignment to action cluster usage profiles, and the histogram of resulting p values is reported.
- (B) Individual behavior usage was calculated for data shuffled as in (A), and the pooled p values across all behaviors are reported.
- (C) p values for (B) following a Benjamini-Hochberg false discovery rate (BHFR) correction.
- (D) p value histogram for a single behavior from the shuffled comparison described in (B) is plotted.
- (E) p values for (D) following a BHFR correction.
- (F) Difference and fold change for action cluster usage for each ASD KO model as compared with WT littermates for lone behavior.
- (G) Difference and fold change for action cluster usage for each pair of KO animals as compared with WT littermates during social interaction.
- (H) Difference and fold change for joint cluster usage for each ASD KO model as compared with WT littermates during social interaction.
- (I) Fraction of time with any contact between fitted deformable mesh bodies during interaction across ASD model strains. Significance is indicated after fitting an LME model for each comparison, $p < 0.05$.